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DEADLINE: 26 MARCH 2025, MONDAY.

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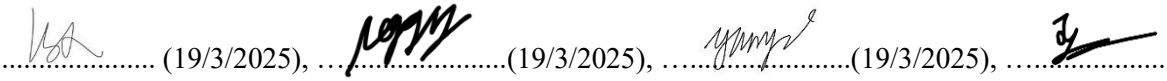
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Academic Honesty Acknowledgement

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1.0 - Introduction

Enterprise Architecture does not have a proper definition around the world. But it can be defined from a business perspective and information technology (IT). Based on a business perspective, it illustrates an organization can achieve their business objectives by creating project initiatives or models that can easily understand (Enterprise Architecture Center of Excellence, n.d.). On the other hand, enterprise architecture for IT perspectives focuses on establishing priorities for initiatives and developing an IT roadmap to know the way to use technology in the organization (ProductPlan, n.d.). The purpose of enterprise architecture is a strategic tool to accelerate the ability of an organization by presenting an approach to finding the gaps between the idea and reality (The Open Group, 2022). This is important to have an Enterprise Architecture because every organization is finding a way to improve their efficiency. Besides that, it provides a clear vision of the organization assets to help reduce complexity and allocate resources more efficiently and lower costs (Ardoq, 2024).

1.1 - Background

Company Buyomart is a convenient store chain that operates in the central region of Malaysia. Over the years, the company has successfully stabilized and grown its business by catering to the everyday needs of the local community. Buyomart offers a wide range of essential daily goods, including groceries, beverages, personal care items, and household products. It also provides exclusive services such as mobile top-ups, bill payments, parcel drop-off points, online shopping payment and printing services, making it a one-stop solution for customers for their quick and convenient shopping needs.

1.2 - Narrative

Since Company Buyomart would like to expand their branch in the southern region, Johor to capture new potential market and increase its competitive edge against other convenience stores chain. They have faced some challenges because their store is fully managed by its headquarters in Kuala Lumpur, where all inventory, sales tracking and customer loyal data are processed. In their previous branches, they have faced an inefficient checkout process, especially during peak hours due to the outdated technology. To solve this problem, we should upgrade Buyomart's Point-of-Sales (POS) system. We also require a cloud-based sales tracking to track sales of multiple branches instantly. Besides, we need to gain competitive advantage by partnering with local delivery apps like Foodpanda, Grab and so on. This is because most of the convenience stores such as Family Mart has already partnered with third-party delivery apps to help them deliver their orders. We also require an efficient centralized inventory system for the new branch due to decentralized inventory and we can avoid excess inventory as well as reduce the risk of product expiration.

2.0 - Main Content

The CSVLOD model of enterprise architecture is used to address the challenges and changes of Company Buyomart. An effective enterprise architecture strategy will be developed based on the CSVLOD model, which includes considerations, visions, outlines, standards, landscapes, and designs. By implementing the CSVLOD model, Company Buyomart may enhance a few aspects, such as inventory distributions, ideal integration with delivery partners, the efficiency of the checkout process, and others.

First and foremost, the important part of the CSVLOD model is considerations. Considerations as a direction for us to decide on some strategies. We need to implement a centralized inventory distribution system, point-of-sale (POS) transactions, and ideal digital connectivity with other delivery platforms.

Furthermore, the standard of the CSVLOD model will be used as a guide towards system architecture and data security. A few guidelines need to be noticed, including retail technology best practices, data protection compliance, and interoperability guidelines. These standards may help Buyomart to implement the systems smoothly.

Besides, the third part of the CSVLOD model is vision. Based on the vision, Buyomart is indeed to develop a retail ecosystem that allows digital integration that can improve the efficiency of operations and customer experience. API-driven framework and an automated inventory management system need to be implemented to enhance stock updates and delivery platforms.

In addition, Buyomart implements landscape and separates it into different domains, which include business processes, technology infrastructure, data management, and others. It describes Buyomart's inventory workflows, checkout processes, inventory analytics, real-time data processing, and others.

After that, the outline of the CSVLOD model requests us to have a centralized POS system and automated checkout solutions. We will also need to use AI to predict customers' demand. It may enhance the efficiency of controlling and delivering stock.

Last but not least, the design part of the CSVLOD model provides layouts for various solutions. For example, the POS system needs to be designed as a cloud-based platform while an automated stock replenishment system needs to be designed for inventory management.

2.1 Discussion Points

The following discussion points are the key points that Buyomart needs to take into consideration while expanding its businesses and improving its systems and operations. The points ensure that all the changes such as upgrading the POS system, collaborating with delivery app services, improving the inventory system, and using cloud-based system tracking are practical and align with Buyomart's business goals. Other than that, these points ensure smooth conversations between different departments and make sure that everyone understands the changes and their impacts on the staff, customers, and overall performance.

2.1.1 Operating model

Buyomart should adapt to the unification operating model, which emphasises on high data integration and high process standardisation. This model suits Buyomart as it operates as a centrally managed organization with standardised business processes across all outlets. The model also ensures all stores follow the same procedures, use the same IT systems, and share data seamlessly. This enables Buyomart to maximise its efficiency, reduce costs, and deliver consistent customer experience.

1. **Business units**

   Buyomart operates primarily in the central part of Malaysia, namely around and across the Selangor state. It has over 128 stores, all of which follow standardised business processes. Each outlet offers a wide range of daily essentials and convenient services such as bill payments and printing. All of its stores are strategically located in residential areas and focus on providing affordable easy access to daily necessities and convenient services. Buyomart focuses on serving local communities by offering convenience, affordability, and consistent quality of services across all outlets.

2. **IT landscape**

   The IT infrastructure of Buyomart has to support its unification operating model which is designed for centralisation and standardisation of its data. The centralised databases ensure that all data is consistent and accessible across all outlets. Such databases allow real-time inventory tracking, sales monitoring, and customer data analysis. The Point-of-Sale (POS) systems of Buyomart needs to be standardised across all outlets to enable seamless transactions during peak hours and consistent customer experiences. Its inventory management also needs to be centralised to track stock levels and optimise supply chain operations. The operating model has to ensure all IT systems can be accessed everywhere so that the headquarters and staff may monitor and manage operations anywhere in real-time.

3. **Key features**

Buyomart's operating model should be built around three key pillars, which are efficiency, cost-effectiveness, and consistent customer experiences. These features can be achieved through streamlined product ranges, economies of scale, efficient distribution of goods, and standardised processes. The mart should focus on offering a curated variation of high-demand items, ensuring that the common needs of the customers are fulfilled. This minimises the complexity of its inventory and eases the purchasing process of customers. The company should also leverage bulk purchasing for lower prices from suppliers. This will in turn result in a more competitive and affordable price range to be provided to its customers. The efficient distribution of goods can be achieved by ensuring a centralised logistics system, which reduces transportation costs and ensures timely delivery to stores. All outlets should also follow the same business processes of Buyomart, which includes inventory management and customer service to ensure consistency and minimal operational variability.

4. **Strategic leverages**

The unification operating model brings strategic advantages that enable the mart to achieve its competitive edges. For example, the standardised processes and centralised systems improve the operational performance and efficiency of the outlets. Buyomart can also control the costs of goods by focusing on bulk purchasing, efficient distribution, and standardised operations to keep costs low, which enables affordable price offerings to customers. It also ensures consistent customer experience by ensuring all outlets follow the same processes and offer the same products and services. The operation model also makes expansions easy as new outlets can be quickly integrated into the system with minimal adjustments. Centralised data systems also enable Buyomart to analyse sales, inventory, and customer behaviour in real-time, which allows informed decision-making and quick adjustments to market trends.

2.1.2 Business Capability

Point-of-Sales system & checkout process

Buyomart faces slow checkouts during peak hours due to outdated Point-of-Sale technology. Customers will have to face a long queue and wait for their turn for checkout. According to Facit (2023), most people will wait no longer than four minutes and will abandon their purchase when it reaches five minutes in the queue. It is also mentioned that purchase abandonment of customers is influenced by the customer's waiting time and transaction inconvenience. Customer retention is important as it is five times harder to acquire a new customer than it is to retain an existing customer. Buyomart currently employs a basic Point-of-Sale (POS) system to run its day-to-day activities. While the system is adequate for its current operations, it is very basic and does not have the advanced features that modern retail businesses should possess. For instance, the POS system is not connected to any inventory management system, and thus the employees have to enter the data manually.

This has the potential to lead to errors like incorrect stock levels which makes it harder to have an idea of what stock is currently available. Furthermore, the checkout process is slow and ineffective especially during promotion seasons and peak hours which in turn adds to the length of lines and poor customer experience. Other than that, the lack of real-time analytics and reporting makes Buyomart unable to identify information about sales trends, customer behaviour, or the overall performance of the business. In short, while the current system is getting the job done, it is holding the business back from growing and staying competitive in today's retail landscape.

Local Delivery App Collaboration

Other than selling goods physically, Buyomart intends to expand its activities online and target remote consumers through third-party delivery app partnerships. As a physical business, Buyomart relies largely on physical buyers to purchase products and services on site. While this model has worked for Buyomart in the past, the lack of a delivery system hinders it from meeting shifting customer needs, especially when convenience and fast delivery are the now major drivers of customer satisfaction. In its absence, Buyomart is unable to tap into the growing demand for home delivery and online shopping. This is not only a restriction to its customer base but also puts Buyomart’s business at a disadvantage in competition with other retailers that offer delivery services. Besides, the absence of delivery services also implies that Buyomart will be unable to apply data from the delivery process towards gaining insights on customer tastes, delivery tendencies, or areas of improvement. Thus, Buyomart is missing potential chances of broadening its horizon, achieving more sales, and securing more interactions with customers.

Improved inventory management

Buyomart suffers from inefficiency in inventory distribution as it is prone to overstocking of products that leads to a high risk of product expiry when the products are not sold in time. This is because Buyomart has a semi-automated inventory management system for monitoring stock levels, controlling replenishments, and tracking product movement in its stores and warehouses. This existing system is prone to human errors, lacks real-time visibility, and is not a part of other business processes such as the POS and sales tracking system. This is most likely to result in stockouts, overstocking, and inefficient supply chain management. The system is not scalable and will be unable to handle the growing complexity of Buyomart's operations with expansion of its lines of products and points of stockholding throughout Malaysia. The lack of real-time data and forecast capability also prohibits Buyomart from being in a position to forecast demands and optimise the levels of stock, resulting in lost sales opportunities and increased costs of holding stock. Although the existing system has the capabilities to conduct necessary inventory managements, it cannot cope with the demands of a modern and data-driven retail setup.

Cloud-based sales tracking

Buyomart employs an outdated tracking system where it provides functions such as capturing in-store sales data, generating daily sales reports, and tracking in-store product availability. The systems are not scalable, lack real-time data processing, and is highly dependent on manual inputs, which leads to delays and errors while reporting data. Moreover, the system is not remotely accessible, therefore it is inconvenient for employees to access the data while away from the office. While the current system operates Buyomart's core functions effectively, it cannot handle the increased volume of transactions and increased complexity of the businesses during peak hours. This will also result in inefficiencies such as slow decision-making, poor customer service, and unable to utilize data for strategic purposes. Besides, being disconnected from modern tools and platforms also hinders Buyomart from innovating and being competitive in the marketplace.

2.1.3 Specific business need

Point-of-Sales system & checkout process

Buyomart should implement a suitable POS system that seamlessly integrates with its inventory management system, sales tracking system, and sales database. A good POS system can reduce the time taken per transaction, which further results in serving more clients and improves potential revenue for stores. Data integration can also provide the management team of Buyomart with a unified view of the mart's health. A good POS system allows the mart to easily manage stocks and maintain price consistency as all data are synchronized in the system (Nguyen, 2024). Other than that, the system should also accept various payment methods including cash, credit and debit cards, e-wallet payments and DuitNow payments to ensure faster and easier transactions.

Buyomart can also invest in self-checkout kiosks with simple, straightforward, and user-friendly interfaces for an alternative checkout option for customers. Other than cutting short checkout queues, customers could get more privacy and control over purchases. Kiosk machines are also cost-effective as it makes it possible to double checkouts and sales without the expenditure of an added employee. It is a long-term investment for Buyomart as the increased sales and decreased labour cost make up for the hefty initial cost of the kiosks.

To further improve the process and identify bottlenecks or any areas for improvement, Buyomart should utilise the data collected by the POS system and automatically synchronise it with the sales tracking system and inventory management system. For example, insights like sales trends, peak hours, and customer behaviour can aid in staffing optimisation, inventory management, and marketing. Performance indicators like average transaction time and customer throughput should also be closely monitored. Feedback is also important for enhancements, short surveys can be done to customers for their opinion on checkout experiences (IT Retail, 2024).

Local Delivery App Collaboration

Collaborating with local delivery apps allows Buyomart to gain a competitive advantage through its convenience and accessibility. Other than food deliveries, delivery apps like Grab and Foodpanda now offer grocery delivery services. Collaborating with such delivery apps could bring benefits such as increased revenue potential, streamlined delivery operations, and boosted brand exposure and marketing (Voiceplug, 2024). Buyomart can be exposed to a new customer base that may not know or have visited the mart. The convenience of ordering through an app can also encourage purchase behaviours of customers without having the trouble to travel to the store. By collaborating with delivery apps, Buyomart gets to utilise the streamlined delivery process they offer. As the platform has an experienced fleet of delivery drivers and systems, Buyomart does not have to use extra effort to hire and manage delivery staff and save time and money in the long run. The mart only has to focus on its existing products and services and allow delivery experts to handle the logistics quickly and efficiently. Seamless order processing can be achieved by connecting the delivery app interface to the inventory management system and POS System for real-time stock availability and payment confirmation. Furthermore, Buyomart can get boosts in brand exposure and marketing by collaborating with delivery apps as the app usually has extensive marketing strategies and a large user base. This in turn benefits Buyomart by increasing its visibility and reach. Buyomart can also offer promotions like free delivery with minimum purchases and vouchers to further attract and drive customers' purchase decisions.

Improved inventory management

To improve its outdated inventory management system, Buyomart can refer to the logistic system of 99 Speedmart as they are known for their low-cost strategies. According to Ng et al. (2018), 99 Speedmart centralise their purchases where purchase orders are issued from the purchasing department in its headquarters. The deliveries will be directed to the distributing centres that serve the designated outlets nationwide. Buyomart can refer to such systems as it enables the mart to get favourable trading terms and discounts by achieving optimum economies of scale via bulk purchases. These actions can ensure a competitive price and low administration costs. To ensure timely restocking to outlets according to sales trends, Buyomart can set up distribution centres and operate them via the logistics team to obtain control in retail planning and merchandise purchasing. Goods arriving on time and in good condition and quality can ensure a competitive edge to the mart. Inventory updates can also be automated through integration with the sales tracking system and POS System, allowing demand forecasting to be more precise. An efficient supply chain which is supported by fully integrated distribution centres working in synergy and an advanced logistic system with efficient transportation allows lower inventory cost, lower size lots, high productivity, more flexibility and faster response time to customers.

Cloud-based sales tracking

Sales tracking is useful for Buyomart as it centralises all data and shows timely insights to the headquarters. Sales tracking enables Buyomart to gain a deeper

understanding of customer behaviour and interests. It allows Buyomart to know about the customer's preferences and enables the mart to properly prepare and execute the best actions. By doing this, it is easy for the mart to gain new customers and maintain relationships with existing ones. The sales tracking system also allows the automation of routine tasks, which frees up the employee's time to focus more on one-to-one interactions with clients and make sure their needs are always entertained (Gradwell, 2020). A sales tracking system can also aid in inventory management and demand forecasting by pulling real-time data from the POS System and inventory management system. It helps in balancing sufficient stock without overstocking and wasting unnecessary capital. By analysing historical sales data, seasonal trends, and external sources like economic conditions or public holidays, the mart can make informed decisions about inventory levels and reorder points. Data-driven demand forecasting also helps retailers plan for promotions, new product launches, and seasonal fluctuations more accurately (Ericksen, n.d.).

The sales tracking system can help in operation optimisation and cost reduction. Through data and industry analysis, the mart can identify inefficiencies in the supply chain, store layout, and workforce management. Buyomart can analyse store traffic patterns and checkout times to allocate employees more efficiently to ensure customers receive prompt service. Other than that, the data can provide competitive advantages by allowing the mart to innovate and adapt to changing customer preferences and market trends. Data analysis can reveal emerging trends and customer preferences that competitors may not be aware of. Being ahead, the mart can introduce new products and services to meet the demands. The innovation can be extended to areas such as marketing strategies, customer engagement, and technology adaptation. Buyomart can utilise the data and experiment with new marketing channels, customer engagement tactics, and emerging experiential retail technologies like AI or augmented reality to enhance shopping experiences and stand out among its competitors.

2.1.4 Business processes

Point-of-Sales system & checkout process

Buyomart should prioritise enhancing its existing checkout process by investing in a modern Point-of-Sales (POS) system that integrates seamlessly with its inventory management, Order Management System (OMS), and centralised Cloud Retail Platform. The existing outdated POS infrastructure contributes to slow transaction times, long queues, and poor customer satisfaction, especially during peak hours. An upgraded POS system will enable faster barcode scanning, reduce transaction time per customer, and thereby increase customer flow. This directly improves the overall shopping experience and revenue potential in the store through increased efficiency and greater customer turnover.

Besides, the latest POS terminal will have to cater to varying types of payment such as cash, credit and debit cards, and e-payments such as e-wallet and DuitNow QR payment.

By processing alternative payments, checkout systems in Buyomart becomes easier to execute and offers reduced lag time as well as a better customer experience. The POS system must also integrate with CRM tools and delivery partner platforms to enable a seamless omnichannel experience.

IT solutions should ensure that the POS system is cloud-based and integrated with Buyomart's central database to allow real-time synchronization of sales inventory and delivery order data. APIs should be used to enable compatibility with multiple payment gateways. The system should also allow remote diagnostics and updates, ensuring minimal downtime and better performance monitoring through centralized dashboards (Razorpay, 2024). From an organizational perspective, a dedicated POS support unit should be established, with employees responsible for system administration, troubleshooting, and cashier training. A Digital Checkout Coordinator role can be introduced to monitor POS system performance analytics and optimize transaction efficiency.

To further modernise operations, Buyomart should also consider deploying self-checkout kiosks with intuitive, user-friendly interfaces. These kiosks not only reduce queue lengths but also empower the customer with greater privacy and control over their purchases. They are beneficial in promoting contactless interactions, especially considering the post-pandemic retail environment. Self-checkout machines, although requiring significant initial investment, offer long-term cost savings by reducing dependency on manual labour while simultaneously doubling checkout capacity.

To implement the new system, the self-checkout system should connect with the POS backend infrastructure to enable real-time synchronisation of sales and payment processing. It should also include security features such as camera monitoring, weigh scales, bag checks, item limits, and alert systems for unpaid or mis-scanned items (SeeChange, 2024). The interface of the kiosk must be touch-friendly and easy to use with multilingual support and accessibility options. POS systems should also send real-time order information to the Order Management System (OMS) for fulfilment when customers choose delivery options at kiosks. Other than that, some cashier roles can be repurposed into customer service assistants for self-checkout support. HR policies must also be revised to accommodate new roles and performance metrics tied to digital transactions.

The POS system also needs to function as a data gathering engine to gather information such as the time of the transaction, the items purchased, the payment methods, and the customer's personal information (Kholod et al., 2024). All such information can be used to make optimum use of staffing, promotional campaigns, and inventory control. The IT group needs to incorporate Business Intelligence tools such as Power BI or Tableau with the POS database so that trends can be automated in visualization. Real-time data streams should also be fed into both the Cloud Retail Platform and OMS dashboards to be accessible to store managers and the headquarters to enable timely decisions based on live metrics. From the organisational perspective, the data analyst unit should analyse POS data and generate actionable reports for each outlet and region. The staff must be also trained to understand and use performance dashboards effectively.

Local Delivery App Collaboration

To maintain its competitive advantage, Buyomart should explore partnerships with local delivery service providers such as Grab and Foodpanda. The absence of delivery services currently places Buyomart at a disadvantage compared to other convenience stores like FamilyMart. Collaborating with established delivery platforms offers multiple advantages, which are wider market reach, higher sales volume, and operational scalability without investing in logistics (Pasirayi et al., 2023). From the IT perspective, Buyomart should develop an API-based integration platform to sync its product catalogue, pricing, and stock availability with partner apps in real-time (Rehman, 2024). A centrally hosted Order Management System (OMS) needs to receive orders from various applications, route them to the appropriate branch, and track fulfilment status in real-time. The OMS needs to be highly integrated with POS and Inventory Management Systems to capture both online and offline transactions in stock balances (IBM, 2021).

Partnerships with food delivery apps can also increase brand visibility via in-app promotions, discounts, and vouchers that encourage high basket size and repeat purchases. Campaign performance monitoring dashboards should be added to the Order Management System to gain insights on which offers perform best. The Customer Relationship Management tools should also be integrated to capture the customers' behaviour from delivery locations for retargeting in the future. The marketing team should also manage campaign planning with delivery platforms, while the customer service team should handle app-based complaints and feedback.

Operationally, Buyomart must maintain updated digital catalogues, optimise backend order processes, and ensure quick preparation of goods for dispatch. This transition to omnichannel retailing aligns with modern consumer preferences. The POS and OMS must be interconnected to reflect digital order deductions in inventory and synchronize sales records across channels. Store tablets or POS terminals should receive immediate order notifications and guide staff through step-by-step processes to reduce errors and improve order accuracy. New roles such as Online Order Fulfilment Specialist and App Sales Analysts should be introduced to ensure the standardisation of workflow SOPs across outlets to manage digital and physical orders concurrently. These roles should work closely with the Digital Checkout Coordinator and the data analyst unit to streamline digital order performance analysis.

Improved inventory management

Upon expansion, Buyomart’s outdated inventory management system becomes inefficient. Stock imbalances, late restocking, and poor demand forecasting can reduce the mart’s profitability. Like 99 Speedmart, Buyomart should implement regionally distributed warehouses to serve nearby outlets more effectively to reduce lead time and optimise logistics. A Warehouse Management System that is integrated with POS and OMS and a Cloud Retail Platform must be capable of tracking inventory levels in stores and distribution centres. RFID and barcode scanning must be employed to track inventory accurately, and the system must provide for stock transfer automation between outlets and distribution

centres (Harvey, 2025). At an organizational level, a regional logistics function needs to be established with distribution centre managers, regional buyers, and supply chain analysts for each major region. Regional and central functions need to be split into two distinct functions. To further improve the operations, Buyomart needs to have AI-based forecasting systems for calculating demand, considering past sales patterns, seasonal variations, weather, and regions to calculate restocking plans (Rafalski, 2024). AI engines need to run the forecasting models directly linked with the procurement module of the Enterprise Resource Planning, producing re-order quantities and calendars.

They need to be automated for outliers like anomalously high-demand spicks or standstills. Forecasting models have to tap into real-time POS and delivery sources of sales data to be as accurate as possible (Vela, 2024). The technology team from the supply chain department must interpret the forecast, adjust procurement plans, and communicate with suppliers on ideal reorder points.

As logistics efficiency depends on integrated operations between suppliers, transport teams, and the outlets, Buyomart should implement a Transport Management System that maps optimised routes and tracks delivery KPIs (SAP, 2023). The management system should integrate GPS tracking, fuel cost analytics, and delivery schedules to enable the headquarters to monitor delivery efficiency and flag delays or route deviations. Roles such as Route Planner, Fleet Coordinator, and Transport Analyst should be added under the regional supply chain teams to manage day-to-day distribution.

Cloud-based sales tracking

Manual sales reporting decelerates decision-making and lacks cross-branch visibility. A cloud-based sales tracking system will enable real-time monitoring of performance and increased cross-branch comparability, thus boosting strategic responsiveness (TYASuite, 2024). POS systems in all stores should automatically report to a centralized Cloud Retail Platform, with view-only access to a role-based dashboard by headquarters, regions, and finance teams. There need to be cloud service service-level agreements that are in place for system uptime (GoodWin, 2024). A Cloud Operations Team may also be set up with Branch Reporting Officers to monitor that every branch is using the system correctly and in a consistent manner. Moreover, analysis of sales data can indicate trends in customer preference, best-sellers, peak seasons, and local behaviour. These insights aids in fine-tuning the inventory, staffing, and marketing. The system should also generate automatic sales and trend reports using BI tools, which is visualises each branch and each product category. Cross-functional real-time analytics need to be in place to enable cross-functional decision-making. Artificial intelligence-driven analytics modules can suggest follow-up steps like price offers or product replacement. The Retail Data Insights Unit will need to supply inputs periodically for the business units and store-level personnel will need to be trained to act on trends through local programs.

Additionally, the cloud platform also supports the automation of routine tasks such as sales reporting to enable staff to focus on value-added services. The CRM integration should trigger personalised SMS and email offers based on customer buying history and

preferences. The Digital Transformation Office should oversee all system integrations, track system usage analytics, coordinate technology upgrades, and ensure interoperability across all platforms. Buyomart should also establish a Digital Transformation Office to lead system rollouts, and training programs, and monitor KPI improvements across all units.

2.1.5 Business requirements

Point-of-Sales system & checkout process

Improvements to the POS system and checkout processes in Buyomart should be done to improve transaction efficiency, reduce checkout times, and provide a more convenient, tech-enabled shopping experience for customers. Due to the evolving customer expectations, traditional cashier-dependent processes are becoming outdated, especially among the younger generations. Therefore, by adopting a robust and scalable POS system, Buyomart can integrate faster checkout options, support diverse payment methods, and ensure seamless synchronisation with its inventory and sales system. Additionally, the implementation of self-checkout kiosks will reduce manpower dependency while increasing customer autonomy, especially during peak hours (Trust Payments, 2024). This transformation will also provide a foundation for omnichannel retail strategies, where customer behaviour and transaction data from both physical and digital touchpoints are unified for business insights.

Functional requirements

From the business perspective, the POS system should act as the central mechanism for driving transaction efficiency and improving the overall customer experience. Its core functional behaviour includes fast transaction processing, automated price application, and real-time inventory deduction after transactions. Cashiers need to be efficient at scanning items, applying promotions and discounts seamlessly, as well as processing varying payment modes including cash, credit and debit cards, e-wallet, and DuitNow QR accurately without delays. It should also enable easy return, exchange, and refund processes seamlessly so that after-sale activities can be handled professionally and confidently by store personnel. Self-checkout kiosks should also operationally allow customers to handle purchases independently while maintaining transaction integrity through barcode scanning and bagging check functions. These kiosks should promote contactless transactions and enable privacy for customers. The POS system should also allow promotion targeting, loyalty program integration, and sales campaign deployment. Data collected from the POS system should allow store managers to expect intuitive dashboards for sales summaries and performance analytics to make immediate decisions on staffing and promotions.

Technically, the POS system must be a digitally integrated system with real-time synchronization between systems like the central database, inventory control system, and payment gateways. The system must perform automated recording of sales, barcode decoding, promotion validation, and multi-mode payment processing, all within seconds of

a transaction to prevent the wasting of time. It should also support application program interfaces (APIs) to integrate with future modules such as delivery applications, customer relationship management (CRM), or business analysis software. On a technical level, the system should support live inventory updates in a manner that when a product sells out, the inventory detail is updated in real-time in all databases. The POS should also support multi-register synchronization so that transaction integrity is supported if one terminal fails. The backend monitoring feature must be incorporated into the self-checkout system in the form of AI-driven fraud detection, video surveillance integration, and item mismatch detection through weighing scales. It must also generate real-time performance diaries, transaction histories, and auto-notifications for unusual actions such as voided transactions that had been inputted multiple times.

Non-functional requirements

From the business perspective, the POS system should reflect reliability, responsiveness, usability, and scalability to enjoy smooth day-to-day functioning and effective customer experience. The system needs to be available with zero downtime to business users in times of heavy business activity or sale. A slow or unstable POS system has a direct impact on lost sales, customer dissatisfaction, and loss of reputation. Hence, a minimum of 99.9% system availability is expected at all the branches. In addition, usability and system ease of use to the employees are also of highest concern. Interfaces must be simple enough to facilitate rapid training of new staff, with self-kiosks designed to offer touch screen input, multilingual guidance, and ease-of-use features like print enlargement or voice output for disabled users. Scalability is also a necessary non-functional requirement because business growth needs the system to accommodate greater terminals, additional stores, or other payment modes without extensive re-engineering. The business also expects that data captured by the POS system is accurate and relevant, with decision-making simplified through visualization tools like Power BI or Tableau.

From the IT perspective, the non-functional requirements emphasize system performance, security, reliability, and maintainability. The POS system should respond to user input within 1–2 seconds for smooth interactions and to prevent transaction pauses. Performance tuning, hardware compatibility, and cloud infrastructure optimization are required to support hundreds of simultaneous transactions without delay or freezing. All financial transactions and customer information must be encrypted in transit and at rest to comply with PCI-DSS compliances (Payment Card Industry Data Security Standard) and ISO 27001 certifications (Kallesen, 2024). Role-based access control, password policies, and automatic session timeout must be applied in the system to prevent misuse or unauthorized access. Additionally, the system should log all user activity for audit trails and provide remote diagnostics, automatic software downloads, and rollback capabilities to minimize operational disruption for upgrades or maintenance (Wang, 2024) The POS system should include failover capabilities, offline transaction modes, and automatic data backups to be usable even in case of temporary connectivity loss. Furthermore, the system should ensure modular scalability, with new features or integrations being added without

impacting core functions. IT staff should develop and maintain centralized monitoring dashboards with real-time insights into system health, usage patterns, and error logs.

Local Delivery app Collaboration

By collaborating with local delivery apps, Buyomart makes convenient shop shopping simpler and more customer friendly. As people increasingly value convenience and on-demand services, Buyomart is moving beyond the need for in-store visits. Customers can now easily shop online and have their items delivered right to their doorstep. Other than that, integration with the store's inventory and POS systems ensures that stock levels are accurate, orders are routed efficiently, and processes runs smoothly. This transformation not only boosts sales but also turns Buyomart into a modern, omnichannel retailer, thus offering simple digital experiences, convenient delivery, and also gain useful insights into what customers want and how well services are performing.

Functional requirements

From a commercial perspective, the collaboration with local delivery apps can push Buyomart beyond storefronts, so customers can purchase products remotely. Customers have to be able to browse an electronic catalogue that mirrors the in-store's offerings in real-time such as product description, price, promotion, and stock status. Orders have to be automatically routed to the respective store based on the location of the customer to enable timely delivery. Besides, the business should also expect the app to support multiple payment methods, including credit and debit cards, e-wallets, and fast payment systems like DuitNow QR, to meet different preferences. Buyomart should also be able to drive promotion campaigns and promotions through the delivery app platform to drive traffic and higher basket sizes. Besides, another basic feature is the order tracking capability, where the customers are provided with real-time status of their deliveries which increases degrees of transparency and faith in the service (Hopstack, 2024). Secondly, the system should also allow for scheduling of delivery so that the customers can choose time slots that are most suitable to them, hence increasing satisfaction and loyalty.

From the IT perspective, the delivery app collaboration must be supported by a robust API-based integration framework that connects Buyomart's internal systems including the Point-of-Sales (POS) system, Inventory Management System, and Order Management System (OMS). The primary functional behaviour includes real-time synchronization of stock availability between Buyomart stores and the delivery platform to avoid issues like overselling or out-of-stock items. Orders placed through the delivery app should automatically be recorded and routed to the closest partner store and tracked by the OMS until order fulfilment. Payment gateways also have to be securely integrated in the system architecture, with API connectors enabling real-time transaction verification and confirmations. The OMS needs to support multiple incoming orders at one time, assign them to respective stores, and provide real-time dashboards for tracking orders by the store staff. The system also needs to be integrated with CRM systems to gather and save customer

buying history and behaviour data for future loyalty program deployment or targeted marketing. It should also be capable of generating automated order receipts, dispatch confirmations, and electronic invoices, based on modern e-commerce standards.

Non-functional requirements

On the business side, the app for delivery must be made available in a uniform, scalable, and friendly way in line with the Buyomart brand image and service level. The uptime of the app must never drop below 99.9% to guarantee uninterrupted service, especially during spikes in demand periods such as public holidays or special promotions. Business owners also require an interactive interface, whereby customers can browse, order, and pay with minimal lag or errors. The system also has to be scalable, to keep up with future growth, whether by product line, more stores, or more delivery areas. Easy access and easy use are also important when trying to serve a large customer base. Customer care must be embedded in the application, either through chatbots or direct interaction with helplines, to advise users in real-time and reduce frustration. Marketing and campaign management capabilities within the system must also allow store managers to monitor campaign effectiveness. For example, they can monitor the discount uptake and sales surges so that they can refine promotional approaches based on real-time responses.

From the IT perspective, critical non-functional requirements for performance, security, interoperability, and maintainability must be met by the delivery app integration. The system should be architected to withstand high concurrency levels under heavy loads without failure or degradation. The loading of the page and acceptance of user inputs must be conducted quickly to enable smooth browsing and transactional experiences. Backend architecture needs to be cloud-native to be able to expand horizontally on services during promotion peak times and automatically adapt to increased loads. End-to-end encryption should be applied to transactions as well as customer data exchanges. The system must comply with local data protection laws such as the Personal Data Protection Act (PDPA) or General Data Protection Regulation (GDPR) and provide support for functionality such as two-factor authentication, role-based authorization, and audit logs. From an integration point of view, the delivery application system should have modular APIs so that third-party operations such as innovative payment providers, marketing platforms, or logistics operations can be supported with simple plugins without destabilizing core operations. IT teams must implement monitoring dashboards, incident response systems, and automated update tools to ensure the platform remains secure, stable, and continuously updated. The system should also support business intelligence tools integration such as Power BI or Tableau to allow the generation of delivery analytics reports, customer purchasing trends, and campaign performance summaries. Regular testing and performance audits must be scheduled to ensure system integrity and disaster recovery protocols should be in place to minimize risks of data loss or downtime. A centralised Cloud-based Order Management System (OMS) should also act as the backbone and integrate all components while providing real-time visibility and control for IT administrators and business managers alike.

Improved inventory management

Buyomart should upgrade its inventory system to ensure smooth operations in its outlets. An improved inventory system management could provide timely stock details anytime and provide accurate forecasted demands based on customer purchasing demands and trend patterns. It will ensure adequate inventory in every outlet and report for the need of restock and purchase from their suppliers. Through an upgraded inventory management system, Buyomart is able to respond to its customer's demands more effectively and further enhances its profits and growth.

Functional requirements

From the business viewpoint, the new inventory management system is the backbone of Buyomart's store operations as it facilitates effective control of stocks, optimises replenishment cycles, and visualises inventory conditions across all store locations. The functional behaviour at the core includes real-time tracking of stocks, automatically triggering low-stock alarms, and reporting the need for re-ordering processes based on specified levels (Schwarz, 2024). The system should also support centralised tracking of inventory so that headquarters and the store managers can view inventory for online channels, warehouses, and all branches simultaneously. This will aid in achieving effective stock allocation and prevents overstocking and understocking, which is unhealthy to profitability and sales of Buyomart. The system should also support product classification, expiry date tracking, batch tracking, and SKU-level granularity to facilitate accurate demand reporting and decision-making (o9 Solutions, 2023). It should also automate the shipping and receipting of goods, such that stock updates are automatically initiated in the system when there a receipt or shipment is received. Employees also require the system to generate inventory performance reports, including inventory turnover ratios, stagnant stock reports, and inventory valuation summaries. These reports can be used to facilitate better purchasing and merchandising plans. Alignment with promotional promotions is also of critical significance because the system needs to update stock quantities as demand increases during promotional periods and prevent over-selling when the items are offered for sale on both online and offline platforms.

On the IT perspective, the improved inventory management system should incorporate real-time synchronization of data to aid in the smooth integration with other systems such as the POS system, Warehouse Management Systems, and supplier order management systems. The automated stock update mechanism which updates inventory counts immediately upon each sale, delivery, or return transaction should be accurate. The system should also support automated purchase order generation based on inventory forecasting models and pre-set replenishment rules to reduce manual workload and improve accuracy. Additionally, the IT system should facilitate multi-store inventory management for tracking between stores or between warehouses and stores. Role-based access control (RBAC) must be implemented in the system so that only authorized personnel will be able to change critical inventory data (Quloi, 2024). Inventory audit trail logs, barcode and RFID capabilities, and data import and export functionality will also be needed by the IT personnel

for stocktaking purposes and system updates. The system should have customer workflows for approval hierarchies for procurement, adjustments, and stock transfers with the ability for the IT administrator to configure processes as business needs.

Non-functional requirements

Businesswise, the upgraded inventory management system must be both reliable and expandable. It should be convenient to utilise with intuitive interfaces and enable easy implementation by all staff from all departments. Business partners expect the system to display the visibility of current inventory information, including dashboards as well as graphical representations which encapsulate complex information into meaningful data. The system must be available with minimal downtime, especially during peak sales periods like holiday promotions or on-sale events. The system must also offer mobile access so that all staff can view or modify stock using hand-held devices or tablets anywhere. Staff members must be able to use the system with minimal training and the system must support multi-language capability and accessibility features to accommodate a diverse workforce. Furthermore, data accuracy and consistency across all platforms are critical for business reporting and strategic decision-making. The system must help reduce the frequency of stock discrepancies and simplify the stock audit process. Business leaders also expect the upgraded inventory system to benefit in lower costs, improved relationships with suppliers, and enhanced customer satisfaction.

From an IT perspective, the improved inventory management system must be in a position to demonstrate high system availability prompt response times, and robust data security controls. It must be working even during peak workloads without loss of performance, with scalability features via modular design or cloud-based hosting. The system must support the integration of third-party plugins or APIs for warehouse automation software, supplier ERP systems, or external logistics providers. Encryption methods, multi-factor authentication, regular backups, and compliance with data protection regulations such as PDPA and GDPR are mandatory. Monitoring dashboards, automatic alerting, and incident response plans must also be implemented by IT administrators to track system health and detect outliers such as spiked demands. It should be capable of offering recovery mechanisms to sustain the business when the system is down or there is data loss. The system should be equipped with version control, modular updates, and sandbox test environments to ensure safe feature rollouts. It should also be capable of migrating data seamlessly in the event of future system upgrades or integrations. The system must also offer the feature to support advanced analytics functionality, such as machine learning-driven demand forecasting and supplier performance, to enable business and IT groups to collaborate on data-driven inventory decisions.

Cloud-based sales tracking

The use of a cloud-based sales tracking system will enable Buyomart to centralise its sales data, achieve end-to-end visibility across all its retail outlets, and make fact-based decisions in real-time. A cloud-based system is more scalable, flexible, and easier to integrate with other business systems. Using the system, sales trends, store performances, customer purchasing behaviour, and promotional efficiency can be monitored at any time from any location through any device. Corporate managers and store managers can utilise real-time dashboards and scheduled reports to enhance price action tuning, promotional campaign optimization, and stock planning. In addition, Buyomart's digitalisation includes sensitive adjustment towards consumer attitude changes, identification of the performance gaps in regions, and increased responsibility for employees. It is also an essential facilitator of omnichannel retail, where balanced online and offline channel sales data are needed in a bid to empower consistent and personal customer experiences.

Functional requirements

On the business perspective, the Cloud-Based Sales Tracking System is an online central system to monitor sales activity throughout all Buyomart stores in real-time. Decision-makers and managers would be able to monitor combined sales data for all the stores in real-time, eliminating delays and inefficiencies of manual sales reporting. Business users require the system to be capable of providing intuitive analytics and dashboards that provide daily, weekly, and monthly insights on sales performance, best-sellers, category-level trends, comparative territorial data, and customers' buying behaviour. The data are required for making timely inventory replenishment decisions, advertisement campaigns, best store performance, and resource utilization. Furthermore, the system must enable reports to be automatically generated, thus keeping manual assembly at bay and enabling planning in advance by the business. The system should be capable of accepting user-defined KPIs and filters to support different business units. The system should also support focusing into individual specific transaction records, analyse sales by the employee, and track promotion effectiveness.

From an IT perspective, the Cloud-Based Sales Tracking System must be provided with real-time automatic synchronization of data from the POS terminals to the cloud infrastructure. The system needs to be integrated with all the related backend systems such as POS system, inventory management system, CRM, and ERP system to ensure standard and proper flow of data within the business ecosystem. The system must offer API integration for seamless interaction with business intelligence software such as Power BI, Tableau, or any other business intelligence system and data sources. IT teams require the system to offer real-time data processing and ingestion functionality to support live dashboard refreshes and low data latency. It should have role-based access controls (RBAC) to restrict access to sensitive sales data by approved personnel based on hierarchy and departmental roles. It should also offer custom report templates, automation of report scheduling, and interactive data exploration capabilities to allow IT administrators to model

reports and dashboards based on business requirements. Furthermore, the IT system must provide audit trails and data export functions to enable business continuity and compliance.

Non-functional requirements

From the business perspective, the cloud-based sales tracking system are required to be usable, scalable, and highly available at all times so that managers can see real-time sales data whenever needed, even after business hours. The system should also be easy to use with easy-to-read dashboards and charts. Additionally, the system should be accessible from a range of devices, from desktops to tablets and mobile phones, for sales tracking on the go. Dashboards and queries should be presented in smoothly and instantly so that they do not create delays during the time of important decision-making. The system has to scale up to accommodate additional users, stores, and more datasets as Buyomart expands. On the front end, the system should be user-friendly for store managers and finance teams to operate without having extensively learn about the operations beforehand. It should also include multilingual support and accessibility features like screen readers or large-font displays to be accessible across employees.

From an IT perspective, the sales tracking system must be cloud-based and have high data security, performance, scalability, and interoperability. It must be familiar with strong data encryption mechanisms, multi-factor authentication (MFA), and regular security audits to protect confidential customer and sales data (Babitz, 2024). The system must comply with the local data protection law compliant such as GDPR, PDPA, and disaster recovery and backup capability to ensure business continuity upon system breakdown or cyber-attacks. Performance-wise, the system must be capable of supporting multiple users at a time and real-time transactions, especially during its peak season. Scalability has to be a built-in attribute, whereby the IT department will be able to scale out new store sites, data pipelines, and user identities without affecting performance. The system must provide monitoring dashboards to IT managers that can track system uptime, latency, user logins, and usage analytics and be monitored in real-time and tuned. Moreover, the sales tracking software must be made modular to allow for easy system upgrades and optimization without data loss or downtime and enhancing scalability and organizational efficiency (Pakkanen et al., 2022). Furthermore, the sales tracking solution must easily integrate with other business applications such as POS, CRM, ERP, and WMS. IT teams should also require tools for automated testing, patching, and version control so that updates can be installed within a controlled and problem-free environment.

2.2 Consideration

Strategic considerations of Buyomart need to be smooth and efficient since we plan to expand the business and operations to the southern region of Malaysia. Considerations are defined as guiding that may affect some business and IT decisions and help them to align with Buyomart goals. Senior business executives and IT architects are responsible for collaborating and developing these artifacts since they will affect critical downstream architectural decisions. Hence, all the IT and business processes should be verified to preconceived principles and policies.

2.2.1 Principles

Principle 1: Centralized Inventory Management Statement: A centralized inventory management system needs to be used by Buyomart to track stock levels in real-time situations at all branches. Rationale: A centralized inventory management system can be useful in terms of ensuring stock optimization, reducing waste, and maintaining the stock level. Implications: A combination of a digital system, automatic restocking alerts, and ideal synchrony with headquarters are required.
Principle 2: Digital Integration with Delivery Platforms Statement: To improve digital sales and last-mile delivery services, Buyomart will have to cooperate with other delivery platforms. Rationale: Various enhancements will be provided to Buyomart through cooperation with other delivery platforms, including expanding the domain of customer reach, enhancing sales potential, and providing competitive advantages. Implications: To reap the benefits of efficient delivery fulfilment, it requires API cooperation with other delivery platforms, rationalized order process, and logistic coordination.
Principle 3: Enhanced Point-of-Sale (POS) System Statement: POS system will be enhanced since it may improve the efficiency of transactions and reduce checkout time. Rationale: A POS system that is contemporary may allow customers to check out faster and easier by reducing waiting time. Implications: Buyomart needs to have enough money to acquire cloud-based POS software, staff training, and affinity with various payment methods.

Principle 4: Data-driven Business Optimization Statement: Real-time data analytics and AI-driven insights should be used by Buyomart to enhance its business, including inventory management, sales, customer experience and others. Rationale: Buyomart may allowed to optimize its stock, implement targeted promotions, and enhance business performance by analysing sales and inventory data. Implications: To improve sales and competitive advantages, Buyomart is required to adjust its pricing strategies and campaigns based on the data output.

Table 2.2.1 *Principles*

First and foremost, centralized inventory management is a tool to help Buyomart control and modify multiple branches' stock levels efficiently. The inventory level across all branches is presently managed by the headquarters, which will cause time waste and low efficiency. Hence, Buyomart can ensure there will be no surplus or shortage of stock through the contraption of a centralized digital inventory system. It may also help Buyomart to reduce waste and enhance customer experience since it may also reduce the probability of stock shortage. The implementation of automatic restocking alerts may also use to predict the customer's demand trends and use the data to control stock levels.

After that, digital integration with delivery partnerships as a term of principle is also important to Buyomart. Since Buyomart lacks an online order and delivery system, we need to cooperate with other delivery platforms. Hence, the customer will be allowed to buy their stuff without hanging out and improving their shopping experience. From the point of view of Buyomart, it may also increase its sales and revenue since it can satisfy the requirements of online customers. One of the requirements in this cooperation is the API connection that links between Buyomart's sales system and delivery platforms. This connection may help to enhance real-time updating according to the order and delivery tracking.

Besides, the enhanced Point-of-Sale System, which is also known as POS system is another tool that can improve the transaction speed and customer satisfaction. Currently, Buyomart is facing challenges such as long checkout times. The outdated POS system is the main factor in this situation. Buyomart needs to change the POS system to a cloud-based system to address this challenge since it may allow speedy transactions, especially during peak periods. Buyomart should be aware that the new POS system needs to support various payment methods due to the trend of e-wallets. The POS system also needs to cooperate with the inventory management system to help Buyomart analyse customers' purchase trends and patterns.

Last but not least, the fourth principle is Data-driven Business Optimization. Buyomart can make decisions, reduce waste, refine stock levels, and enhance customer experience through the integration of data analytics and POS systems. The cloud-based sales tracking system can help Buyomart keep track of all the purchasing trends, sales, and stock changes. Hence, it can ensure there will not be a situation such as a stock surplus or shortage that may reduce revenue. Data-driven Business Optimization also offers automatic reports that may reduce the requirement for manual data processing. It may improve overall business performance.

2.2.2 Policies

External	**POS and Checkout Policies**	**Policy 1: Speedy and secure POS System** **Description:** The implementation of a POS system must allow the response time of barcode scanning to remain below 1 second, integrate with various payment methods, and align with cloud-based inventory tracking.
	Data Protection Policies	**Policy 2: Safe Customer Transaction Data** **Description:** To avoid fraud and unauthorized transactions, payment data need to be encrypted and abide by PCI DSS standards.
Internal	**Inventory Management Policies**	**Policy 3: AI-Powered Inventory Optimization** **Description:** The AI-based inventory tracking system needs to be applied by Buyomart to predict demand, avert stock shortage or surplus, and reduce the risk of product expiration.
	Local Delivery Integration Policies	**Policy 4: Secure API Integration with Local Delivery Apps** **Description:** To allow real-time order processing and tracking, Buyomart is required to cooperate closely with local delivery apps.
	Sales Tracking Policies	**Policy 5: Cloud-Based Real-Time Sales Monitoring** **Description:** A centralized cloud-based sales tracking may help Buyomart in terms of providing

		real-time purchase patterns, stock levels, and customer purchasing behaviour.
	Customer Loyalty and Promotions Policies	**Policy 6: Automatic Loyalty & Promotions System** **Description:** This system is required to enhance customer satisfaction through offering reward points, personalized offers and free delivery.

Table 2.2.2 *Policies*

First and foremost, a POS system that is speedy and safe is an important tool in terms of reducing transaction time. It can also combine numerous payment methods, which can enhance the efficiency of the checkout process. Hence, the POS system may bring about the reduction of checkout time by approximately 30% and self-checkout that can reduce at least 20% of processing, which leads to minimizing queue.

Furthermore, safe customer transaction data, the second policy is important in terms of protecting payment data through implementing PCI DSS obedience and encryption. This may lead to the avoidance of cheating, data contravention, and unauthorized access which may reduce the customer experience and trust.

Besides, AI-powered inventory optimization is the third policy of Buyomart. It can reduce stock waste and certify the availability of products through analysing demand patterns across all branches and predicting stock. It may also reduce inventory tasks that are processed manually. Thus, it will allow Buyomart's employees to put more effort into customer service.

Moreover, secure API integration with local delivery apps is important in terms of the policies. Buyomart has been allowed to have flawless order fulfilment since the API integration will link Buyomart's ordering system and local delivery apps together. This function will also allow customers to track their orders in real-time. It will also increase the probability of on-time delivery which may enhance customer.

In addition, the fifth policy that can allow the headquarters to monitor the sales of all branches is cloud-based real-time sales monitoring. This system can help Buyomart to automatically produce sales reports. Thus, this policy may enhance product stocking, promotional strategies, and pricing adjustments.

Last but not least, the automated loyalty and promotions system is a system that can focus on customer retention since it can offer rewards and promotions to the customers. Buyomart is expecting to have an increase of 15% in repeat customers through this system. It can also help to ensure that promotions such as free delivery may encourage long-term loyalty and sales in the long run.

2.3 Visions

2.3.1 Business Capability Models

Business Strategy and Vision		
Goal 1: Enhance Operational Efficiency	**Goal 2:** Improve Customer Experience and Engagement	**Goal 3:** Strengthen Data-Driven Decision Making

Capability 1		**Capability 2**	
Capability 1.1: Real-time stock Tracking	**Capability 1.2:** Automatic Replenishment	**Capability 2.1:** Barcode Scanning Efficiency	**Capability 2.2:** Various Payment Method Integration
Capability 1.3: Demand Forecasting	**Capability 1.4:** Supplier Cooperation	**Capability 2.3:** Self-Checkout Kiosk	**Capability 2.4:** Cheat Prevention Mechanisms
Capability 3		**Capability 4**	
Capability 3.1: Personalized Promotions	**Capability 3.2:** Loyalty Program Implementation	**Capability 4.1:** Cloud-Based Sales Tracking	**Capability 4.2:** AI-Powered Business Insights
Capability 3.3: Customer Feedback Analytics	**Capability 3.4:** Targeted Marketing Campaigns	**Capability 4.3:** Real-time Performance Dashboards	**Capability 4.4:** Data-Driven Decision Making

Regular: Basic business functions that can maintain daily operations	**Important:** Important contribution to the business performance	**Strategic:** Provide competitive advantage and align with long-term goals through high-priority capabilities

Table 2.3.1 *Business Capability Models*

Three core goals are the fundamentals of business strategy and vision which are enhancing operational efficiency, improving customer experience, and strengthening data-driven decision making. These goals can be used to help Buyomart maintain competitive and consistently improve its processes. Thus, this may help Buyomart to improve customer satisfaction and industry standards. A structured business capability model has been developed to meet these goals. This can help to assign different capabilities into three different categories according to their importance and effect on business performance.

First and foremost, the first goal of Buyomart is to optimize operational efficiency, which will be concentrated on enhancing internal processes, improving efficiencies, and making business operations easier. A strong inventory management system is very important since it can reduce stock waste and ensure product accessibility. Besides, POS systems can be improved in many ways such as implementing quick barcode scanning, supporting various payment methods, and self-checkout kiosks. Checkout time also can be reduced by improving the POS system. Thus, it can improve the overall customer shopping experience. Buyomart will have smooth operations that can reduce costs and increase revenue through executing these measures.

Furthermore, the second goal of Buyomart is to enhance customer experience and engagement which is focused on providing an ideal personalized shopping experience. A few ways need to be focused on by Buyomart such as incorporating customer loyalty programs, personalized programs, and real-time feedback mechanisms. These may help to build long-term relationships between Buyomart and its customers. Since customers may enjoy some rewards and offers, customer retention will increase. The ordering system may also provide convenience to customers since they can place and track their orders in real-time.

Moreover, the third goal is strengthening data-driven decision-making, which is also an important goal for Buyomart in terms of long-term business sustainability. Buyomart is allowed to figure out the purchase pattern, customer behaviours, and operational performance by using a cloud-based sales tracking system with AI. It can also help to produce reports automatically, track trends, and forecast customers' demands. Hence, it can help to decrease manual reporting, and the management team can put their effort into strategic growth. Inventory management with predictive analytics can also optimize the stock level by avoiding stock surplus or shortage. Buyomart is encouraged to use the data effectively to improve its efficiency and competitive advantage.

Capabilities have been allocated to different categories based on their effect on business operations. There will be three different categories which are regular, important, and strategic. First of all, regular capabilities are important in terms of sustaining day-to-day operations, including inventory tracking, sales processing, elemental reporting functions, and others. Besides, the second category, 'important' is more on advanced functions. This may include automatic inventory refill, fraud avoidance mechanisms, personalized marketing strategies, and others. Hence, the efficiency of processing and customer engagement will be enhanced due to this category. The strategic capabilities as the last category are the most important with the highest priority functions. It may provide Buyomart competitive advantages through AI-driven analytics, decentralized inventory management,

predictive demand in the future, and others. These may allow Buyomart to give attention to marketing trends and satisfy customer needs.

2.3.2 Roadmaps

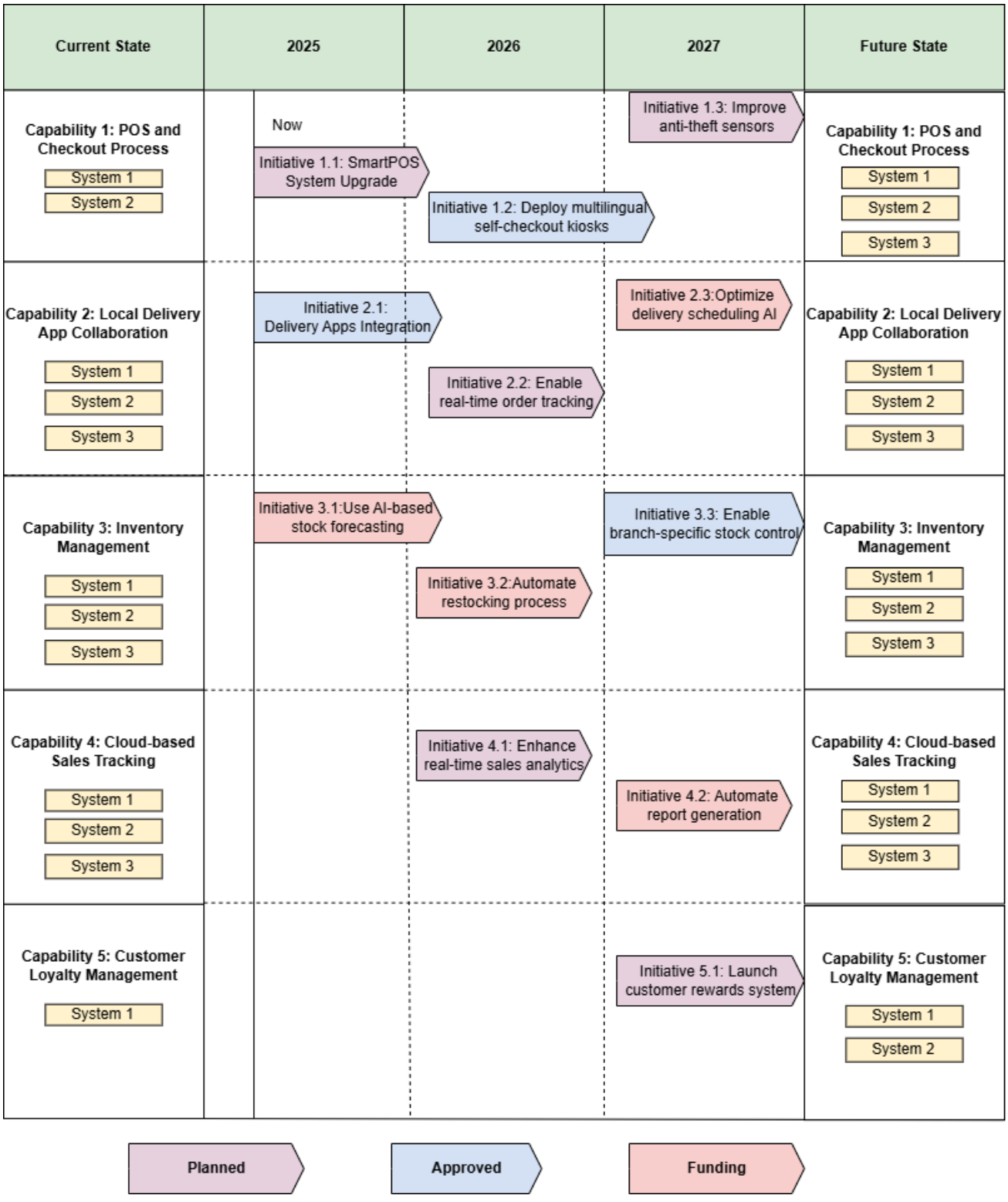


Diagram 2.3.2 *Roadmap*

This roadmap is useful in terms of describing four core operational areas of the system expansion strategy. These areas include a Point-of-sale (POS) system, integration between other delivery platforms, inventory management, and cloud-based sales analytics. Besides, the loyalty rewards hub also joined this system as a new initiative that can help to engage and attract customers.

First of all, we decided to enhance the systems toward digital transactions in 2025. We made some changes to the SmartPOS System so it can reduce the checkout processing time and ensure transaction safety by deploying two systems. Besides, the express delivery integrator will help Buyomart to clinch cooperation with delivery partners by launching one system. The AI stock optimizer will be also used as a tool to help refill stock automatically with the implementation of two systems. Hence, it can also help Buyomart to reduce stock waste. However, cloud-based sales tracking will be used to track real-time sales reports by establishing one system.

In the next year, 2026, Buyomart will expand its investments in the technology domain. For instance, Buyomart will expand another system on the SmartPOS system. Hence, it can bring about AI-power billing and self-checkout that can reduce processing time. The Express Delivery Integrator which has upgraded to two systems may also lead to speedy order tracking and delivery. To enable demand forecasting, the AI-stock optimizer will be upgraded to three systems to provide predictive analysis. Other than that, Buyomart will add another system to the cloud-based sales tracking system to provide convenience to headquarters in terms of monitoring sales systems across all branches.

For the next year, which is 2027, Buyomart will be focused on clarifying automation and AI implementation. The first capability, which is the SmartPOS system, will continue using three systems, but it will have a combination of facial recognition and personalized shopping. Nevertheless, the AI-stock optimizer and cloud-based sales tracking will be expanded to three systems. Thus, they will help Buyomart to have decision-making on stock and sales based on the data analysis output. Besides, Buyomart will launch a loyalty rewards hub with one system at the beginning. This is a capability that can used to initiate personalized promotions and reward programs to customers.

Last but not least, Buyomart aims to have a fully integrated and extremely computerized environment in the future state. Some capabilities, including the SmartPOS system, local delivery app collaboration, and cloud-based sales tracking will control 3 systems for each of them. Hence, it will ensure Buyomart enjoys cost-effectiveness and controllable complexity. However, the loyalty rewards hub will be upgraded to two systems, which can offer more functions, including deeper customer insights and gamification features that can improve customer engagement.

2.4 Standard

Based on Kotusev (2019), standard are IT-focused Rules artifacts that describe global technical rules, standard and best practices relevant for IT system. It developed by architects and technical subject matter experts. The subtype of standards are technology reference models, guidelines, patterns, IT principles and logical data models.

2.4.1 Technology Reference Model

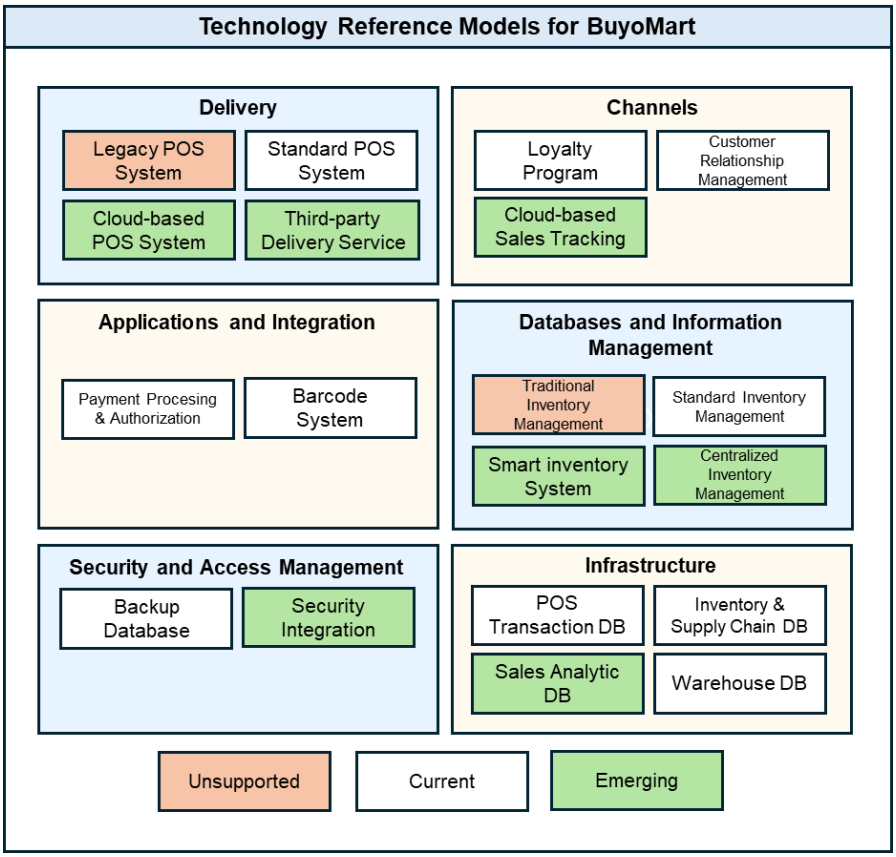


Diagram 2.4.0 *Technology Reference Models*

1. **Delivery**

   Buyomart's delivery systems have replace the legacy POS setups to the current Standard POS system. For now, we planned to upgrade again by replace a new cloud-based POS System. This help to enhance transaction speed and scalability as the employee can access the POS system data in anywhere. Besides that, the third-party delivery services are set to improve customer convenience and increase their satisfaction and loyalty. This will help Buyomart to increase the sales. Thus, the advancement will help in decision making, accessing data quickly and increase the profit as we can have more advance service to our customer.

2. **Channels**

   Customer engagement plays a crucial role to build long-term relationships with our customers. The thing Buyomart have do is by using existing loyalty program and

customer relationship management where we let the customer to collect point to redeem more voucher, bundle and personalized reward. Besides that, we would like to add a cloud-based sales tracking system that offer real-time insights to the customer behaviour and purchasing trends to enable more accurate personalized marketing strategies to optimize the promotion and the customer retention to buy from Buyomart.

3. Applications and Integration

Buyomart transaction infrastructure remain to rely on secure payment processing and barcode system for sales and inventory tracking. The future enhancements may involve RFID tracking and advanced fraud prevention mechanisms that yet to plan from the business team.

4. Database and Information Management

Before we had used the standard inventory management, we have use the traditional inventory management that do not provide real-time update on the inventory from each branch. This cause the inventory will be out of stock. Thus, we have used the standard inventory management that provides real-time update on it. But in the future, we would like to have an AI-powered system that give a predictive analytics and real-time tracking to the inventory. The standard inventory management will remain in the place but emerging with the smart centralized inventory systems. This help to optimize the stock levels and reduce the waste by ensuring the smooth supply chain operations. This help Buyomart to ensure a better coordination of the inventory and lead to improved efficiency and cost saving.

5. Security and Access Management

Buyomart need strengthening its security framework as we going to growth the market in Malaysia. Thus, an advanced authentication and encryption protocols is needed. While we have the backup database to prevent data loss, emerging the security integration measure will provide an advanced protection against the cyber threats. Multi-factor authentication, anomaly detection, and compliance with industry standards will help safeguard customer and business data.

6. Infrastructure

The existing POS transaction and inventory databases handle daily operations, but the introduction of a sales analytics database will enable deeper insights into sales performance, customer trends, and inventory movement. While these three databases are sync and transform to the Warehouse Database. These data will

support and generate a better forecasting, strategic planning, and overall business growth.

2.4.2 Guideline

Guidelines for Buyomart		
POS and Checkout Standards	**Guideline 1**	**Implement cloud-based POS Systems**
	Description	Replace the standard POS system to a cloud-based system for better system reliability and real-time data access in any location.
	Guideline 2	**Ensure POS system response quickly**
	Description	Optimize the product barcode scanning system to speed up the checkout experience.
Inventory Management Standards	**Guideline 3**	**Adopt centralized Inventory Management**
	Description	Use the centralized inventory systems to maintain real-time inventory updates across all branches.
	Guideline 4	**Utilize AI-Based Inventory Optimization**
	Description	Use AI to predict the demand and have restock alert to minimize stock shortages.
Security and Access Management Standards	**Guideline 5**	**Secure API Integration with Delivery Platforms**
	Description	Ensure encrypted API connections with third-party delivery services to enable seamless and secure order fulfilment.
	Guideline 6	**Implement 256-bit Encryption for Transactions**
	Description	Encrypt customer payment data and prevent data breaches.
Sales Tracking and Customer Engagement Standards	**Guideline 7**	**Cloud-Based Real-Time Sales Monitoring**
	Description	Use cloud analytics tools to track purchase trends, stock levels, and customer behaviour in real time.
	Guideline 8	**Loyalty and Promotion Systems**
	Description	Provide more mystery reward programs to personalize promotions and enhance customer loyalty.
	Guideline 9	**Standardize POS Transaction and Sales Analytics Databases**

Infrastructure and Database Standards	**Description**	To ensure seamless data flow between POS systems and analytical platforms for decision-making.
	Guideline 10	**Integrate Inventory and Warehouse Databases**
	Description	To ensure real-time synchronization between inventory management systems and the centralized data warehouse to get accurate predictive analytics.

1. **POS and Checkout Standards**

   To enhance customer experience and operational efficiency. Buyomart is transitioning from legacy POS systems to cloud-based solutions. This change allows for real-time transaction processing, remote access, and system scalability. And we need to maintain a barcode scanning response time quickly to ensure faster checkouts and minimize the waiting time for customer. By integrating these, Buyomart can optimize the transactions while maintaining data security and compliance with industry standards.

2. **Inventory Management Standards**

   Effective inventory management is crucial to maintain optimal inventory levels and reduce the waste. By using the centralized inventory management system, Buyomart can synchronize the stock level from all branches so we can improve the supply chain efficiency. Besides that, the AI-based inventory optimization further enhances the stock management as it help to predict the demand and prevent shortage and automate the restock processes. These standards ensure that products remains available and minimize the cost and spoilage.

3. **Security and Access Management Standards**

   As people are using more digital transactions like card payment and QR payment, the secure API connection is important. Buyomart use encrypted communication between system to prevent any unauthorized access and data leaks. Additionally, 256-bit encryption help to protect customer payment data from cyber threat. These actions will further enhance the security across Buyomart's technology ecosystem.

4. **Sales Tracking and Customer Engagement Standards**

To maintain a competitive edge, Buyomart will introduce a cloud-based real-time sales monitoring to allow branch managers to track purchase patterns, stock levels, and customer behaviour in real time. This approach will help to optimize the product placements, manage the promotions and forecast the customer demand. Additionally, an automated loyalty and promotion system will personalize customer rewards by offering targeted discount and personalized deals based on their behaviour. By using this, Buyomart can improve customer retention and increase the sales.

5. Infrastructure and Database Standards

A good infrastructure will ensure seamless data flow and operational consistency. Firstly, we need to standardize POS transaction and sales analytics databases to enables accurate reporting, data-driven decision-making, and streamlined simplified of financial records. Additionally, we need to integrate inventory and warehouse databases to allows automated stock updates between branch and distribution centres to ensure optimal inventory levels at all locations. These standards enhance the supply chain efficiency, reduced operational risks, and provide better customer service.

2.5 Landscape

Based on the research from Kotusev (2019), Landscape are IT-Focused Structure Enterprise Architecture artifacts that provide high-level technical descriptions of the organizational IT landscape. It developed by architects and maintained current as the IT landscape changes. It used to support technical decision-making and facilitate the planning for Outline and Designs for specific IT project.

2.5.1 Landscape Diagram

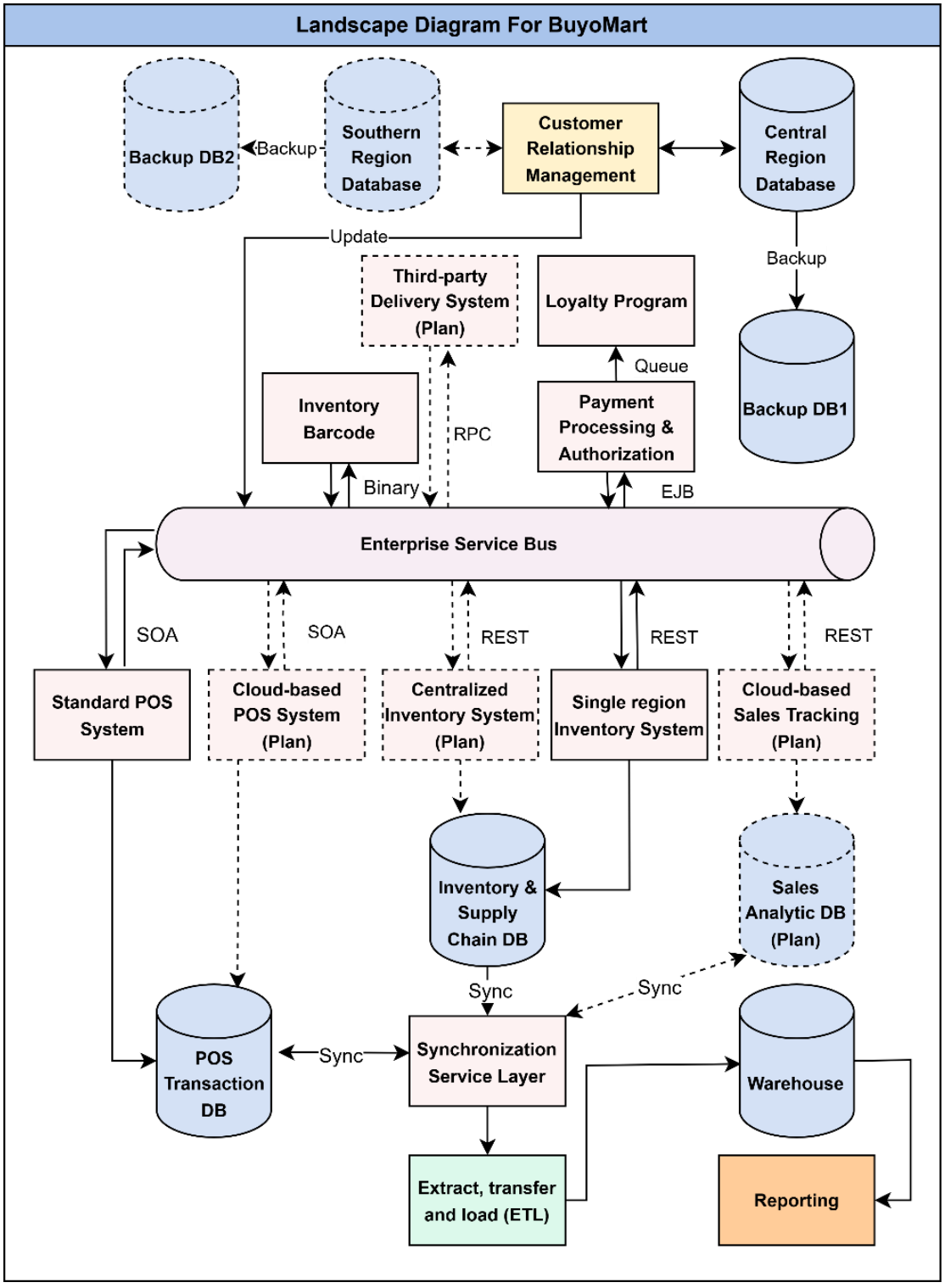


Diagram 2.5.0 *Landscape Diagram*

The Buyomart enterprise architecture landscape diagram **(Diagram 2.5.0)** is designed to support the expansion from central region to the southern region to ensure seamless operation between two areas. It illustrates the way system and database help to ensure a smooth business operations and the plan to improve in the short-term period.

Customer relationship management **(CRM)** Is one of the cores of the diagram where it records customer loyalty program data, including profiles, point and purchase histories. This system is linked to the central region database and the planned southern region database where the sales, inventory and customer transactions from there are processed there. To prevent any data loss or attack, we have backup databases to retrieve data quicky to avoid more business loss.

The other key component in this system is the Enterprise Service Bus **(ESB)**. This is a software architecture that supports communication between different applications and services in an enterprise (Kanade, 2023). It acts as a middleware to connect almost all item in the diagram. This ensures a seamless data exchange between all application such as POS system, inventory system, payment processing system, the planned third-party delivery system and the planned cloud-based sales tracking by different integration method.

The current assets that Buyomart have is inventory barcode system via binary protocol. Binary protocol improves the data transfer speed and efficiency (Databento, n.d.). By using binary protocol, the system can quickly get the price and product information by scanning barcodes on products, parcels and top-up cards. Besides that, the payment processing and authorization system had used the Enterprise Jafa Bean **(EJB)** to connect with ESB that help to manage order payment, inventory validation and receipt. EJB is a framework and development model for server-side components in Java. (Warren, 1999). It is a component-based server-side architecture for creating and managing enterprise-level applications' business logic (Parashar, 2023).

Currently, a standard POS system and single region inventory system are in use but need to be improve as we expand into the Southern Region. We plan to make some several upgrade on the system we had. Firstly, a new database for the Southern region will be introduced to handle sales and inventory for the new branch. It is also supported by a different backup database to prevent data mix with the Central region data. Moreover, a cloud-based POS system will replace the current POS system as it allows admin to access data from both central and southern regions at the same time and different locations. In addition. It will connect ESB by using Service-Oriented Architecture (SOA), a software development model that makes services reusable and lets them communicate across different platforms and languages to form new applications (Barney & Nolle, 2023). In

addition, the centralized inventory management system are planned as Buyomart will have 2 different region data now, with each region have their own single region inventory management system. It will connect to ESB through representational state transfer (REST). REST is an architectural style to provide standards between computer systems on the web to make it easier for systems to communicate with each other (Codecademy, n.d.). By using that, it can manage all branch stock availability using the REST API to fetch live data. Finally, a new cloud-based sales tracking application will be installed to provide a real vision of the sales tracking to both regions. This will help to improve the Buyomart business strategy by analysing the data we have.

Furthermore, a Synchronization Service Layer is needed to connect multiple databases to ensure seamless data exchange between the POS transaction database, inventory and supply Chain Database, and the planned sales analytics database. It helps in exchanging data with one or more external commerce systems, such as databases (Sitecore, n.d.). Thus, it can facilitate real-time updates by synchronizing data across different systems, ensuring accurate inventory tracking and sales records for both the Central and Southern Regions.

The synchronized data is then processed by the Extract, Transfer, and Load (ETL) System. It is also a critical component for BuyoMart's data integration strategy. The ETL System will collect, transform, and load data from multiple sources and load the data to the warehouse database. It plays a crucial role to store aggregated sales, inventory, and transactional records to help enabling efficient analysis of business trends. Finally, the processed data will go for reporting to provide insights into sales performance, inventory levels, and customer purchasing behaviours. These reports will help Buyomart refine its business strategies, optimize inventory management, and improve decision-making across both regions.

In conclusion, this landscape diagram ensures that the expansion for Buyomart is scalable, reliable and efficient by allowing real-time data synchronization, optimized inventory management and enhanced the customer experience. The third-party delivery system also can maintain a seamless process and increase customer satisfaction to keep using the delivery option if they not free to come our convenience store. In addition, the centralized inventory database help to increase the efficiency and cost-saving in checking the inventory balance they had. In addition, the cloud-based sales reporting will record the real-time sales data from other system like database and POS system to generate the sales analytic report to generate the more detailed sales performance reports at the end. In short, Buyomart can maintain competitive advantages, improve the overall operational efficiency and increase the customer satisfaction.

2.5.2 IT Roadmaps

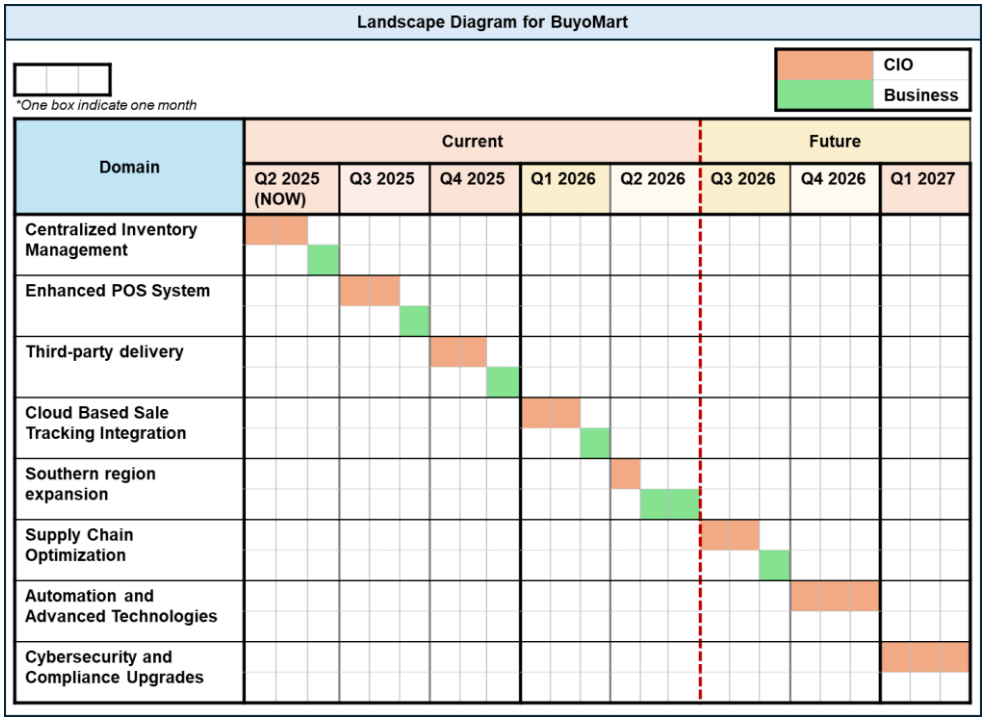


Diagram 2.5.1 *IT Roadmaps*

The diagram 2.5.1 show the IT roadmaps for the Buyomart. In the short-term period (1 year), we need to start from the improvement that we currently wanted to achieve. Firstly, we will proceed to centralized inventory management as it plays a crucial role to a convenient store. This help to improve the stock control and warehouse efficiency by implement the synchronization service layer from the IT perspective in 2 months. Another 1 month is for business team to train employees on real-time stock tracking. In Q3 2025, we will enhance our current POS system to a new one. The IT team will deploy a new cloud-based POS that is more efficiency and cost savings to the current one. The business team will integrate the better loyalty programs and learn to do sales analytics from new POS system. In Q4 2025, we will start to partner with the delivery company such as Grab and FoodPanda for the instant delivery or make our own delivery service. In business perspective, it helps to increase customer experience. In Q1 2026, we will proceed to introduce new system, cloud-based sale tracking. This will help us to enable real-time sales tracking and forecasting on Buyomart as we can get the data anywhere with internet. After these four upgrades is done, we will start to do the Southern region business expansion by opening few branches in Johor and Melaka.

In the future, Buyomart is expected to optimize their supply chain management system to enhance procurement and distribution network by adding AI-driven demand forecasting to improve the supplier relationship management. Besides that, we expected to implement automation in business operation. Moreover, we wanted to strengthen the IT infrastructure by implement multi-factor authentication (MFA). MFA is a way of confirming the real user are trying to log in based on the application such as Microsoft Authenticator

(Microsoft Support, n.d.). This ensures that the authorized person to access to the IT infrastructure.

2.6 Outlines

Based on Kotusev (2016), outlines are business-focused changes which provide business leaders with understandable high-level descriptions of specific IT projects. The aim of developing outlines is to estimate the overall business value of specific IT projects. Architects and business leaders will collaborate to develop the outlines, so that it can be used to evaluate, approve and fund specific IT projects. The examples of EA artifacts related to Outlines are solution overview, options assessments, initiative proposals and so on.

2.6.1 Option Assessments

Upgrading a Retail Point-of-Sales System & Checkout Process

Options Assessments

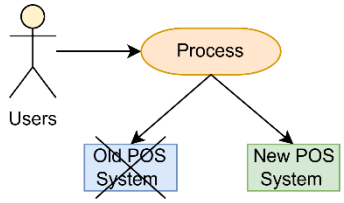
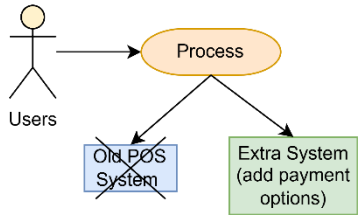
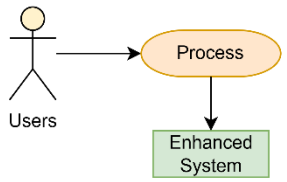
Option	Score
Solution 1:  Time: 8 - 12 months Cost: RM 2 - 3 million Advantages: Faster transaction speeds, real-time sales tracking Disadvantages: Requires training for staff, high cost Risks: Possible system integration issues	Functionality: 5 Feasibility: 2 Alignment: 3 **Total Score: 10**
Solution 2:  Time: 4 - 6 months Cost: RM 1.0 - 1.7 million Advantages: increases transaction flexibility, improves convenience Disadvantages: might have additional transaction fees, does not significantly reduce checkout time Risks: security risk, potential incompatibility with some banks	Functionality: 3 Feasibility: 3 Alignment: 1 **Total Score: 7**
Solution 3:  Time: 3 - 5 months Cost: RM 0.7 - 1.3 million Advantages: faster checkout process during peak hours, reduces dependency on cashiers Disadvantages: potential for theft, high setup cost Risks: maintenance costs, risk of technical failures	Functionality: 2 Feasibility: 5 Alignment: 2 **Total Score: 9**

Diagram 2.6.0 *Option Assessment: Upgrading a Retail Point-of-Sales System and Checkout Process*

Diagram 2.6.0 shows the option assessment of upgrading a retail Point-of-Sales system and checkout process. This option assessment is to provide suggestions for the challenge of upgrading Buyomart's POS system. There are three solutions suggested for these changes. The first one is to completely replace the old POS system into a new one, by including quick barcode scanning, automated pricing and discount function, multiple payment options and enhanced payment authorization. The time required to implement this solution is around 8 to 12 months with an estimated cost of RM2 million to RM3 million. The advantage of this solution is to speed up transaction, as it supports quick barcode scanning with a response time of less than 1 second per scan. Besides, the solution supports real-time sales tracking as the transaction details will be uploaded into the Cloud-based transaction database immediately, which synchronizes with other branches. However, it might be a disadvantage due to high cost of purchasing new POS hardware, which includes terminals, scanners and so on. The employees also require training to adopt the new POS system. The risk of implementing this solution is we will face possible integration issues such as delay, system failures or data mismatches.

The second solution suggested is to only add an extra system which supports more payment options. The time required to implement this solution is around 4 to 6 months with a cost of RM1 million to RM1.7 million. The advantage of implementing this system is it enhances the convenience of customers through adding multiple payment options. However, it might be a disadvantage when it does not significantly reduce checkout time of customers, and it might have additional transaction fees. The risks of implementing this solution are security risk and potential incompatibility with some banks.

The third suggested solution is to enhance the system by introducing self-checkout kiosks, where customers can scan and pay for their items without the cashier's assistance. The time needed to implement this solution is 3 to 5 months with a cost of RM0.7 million to RM1.3 million. The advantages of this system are to speed up checkout process of customers during peak hours and reduce dependency on cashiers. The disadvantage is high setup cost and there might be a potential for theft. The risks of this solution are high maintenance costs and technical failures, such as software glitches.

In this option assessment, solution 1, replacing old POS system into new POS system has obtained the highest total score of 10. Followed by solution 3, add an extra system with a total score of 9 and solution 2, introducing self-checkout kiosks with a total score of 7.

Local Delivery App Collaboration

Options Assessments

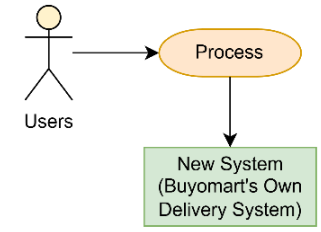
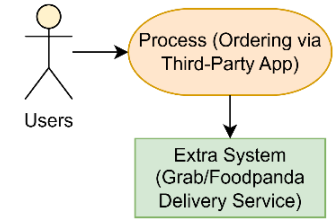
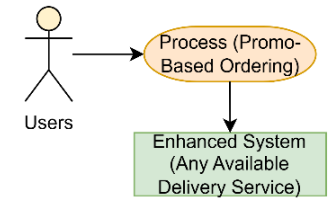
Option		Score
Solution 1: 	Time: 9 - 14 months Cost: RM 2.5 - 4 million Advantages: direct customer engagement, brand strengthening Disadvantages: high development cost, requires ongoing maintenance Risks: technical failure, security risks	Functionality: 5 Feasibility: 2 Alignment: 5 **Total Score: 12**
Solution 2: 	Time: 4 - 7 months Cost: RM 0.8 - 1.5 million Advantages: Faster implementation, lower development cost Disadvantages: Platform commission fees, less control over customer data Risks: changes in platform policies, potential high commission rates which reduces profits	Functionality: 4 Feasibility: 5 Alignment: 4 **Total Score: 13**
Solution 3: 	Time: 2 - 4 months Cost: RM 0.3 - 0.8 million Advantages: quick implementation, increase online sales Disadvantages: temporary impact, potential cost increase if not managed well Risks: reduced profit margins, customers might expect ongoing discounts	Functionality: 3 Feasibility: 5 Alignment: 3 **Total Score: 11**

Diagram 2.6.1 *Option Assessment: Local Delivery App Collaboration*

Diagram 2.6.1 shows the option assessment of local delivery app collaboration. This option assessment is to provide suggestions for the change in partnering with local delivery apps. The first solution is to introduce Buyomart's Own Delivery System, where Buyomart will hire delivery men to deliver the orders. Customers who place orders with a minimum amount of RM30 will receive a RM 10 discount on the delivery fee. The time required to implement this solution is 9 to 14 months with a cost of RM2.5 million to RM4 million. The advantages of this solution are it can strengthen our brand and encourage direct customer engagement. The disadvantages are it requires high development costs and ongoing maintenance of the system. The risks are technical failure, like system crashes and security risks like unauthorized access.

The second solution is Buyomart can collaborate with third party delivery apps so that customers can order via extra systems such as Foodpanda or Grab. The time required to implement this solution is 4 to 7 months with a cost of RM0.8 million to RM1.5 million. The advantage of this solution is it can lower the development cost of Buyomart and it can be implemented faster. The solution can be a disadvantage as Buyomart will have less control over customer's data and the third-party delivery platforms will issue commission fees. The risks of this solution are it might face changes in platform policies and the third-party delivery apps will issue potential high commission rates which reduces Buyomart's profits.

The third solution is Buyomart can implement a promo-based ordering where customers will often receive discounts and promotions while they make their purchase. The time needed to implement this solution is 2 to 4 months with a cost of RM0.3 million to RM0.8 million. The advantages of this solution are it can increase the online sales, and this solution can be implemented in a short period of time. The disadvantage of this solution is the potential cost will increase if not managed well. These discounts and promotions only have temporary impact in increasing the sales and competitive advantage of Buyomart. The risks in this solution are Buyomart will face reduced profit margins and customers might expect ongoing discounts.

In this option assessment, solution 2, partnering with third-party app has the highest total score of 13, followed by solution 1, introducing Buyomart's Own delivery system with a total score of 12. Solution 3, enhancing the system through implementing promo-based ordering has the lowest total score of 11.

Efficient Inventory Distribution and Centralized Inventory

Options Assessments

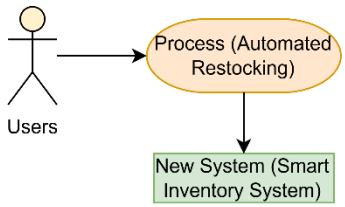
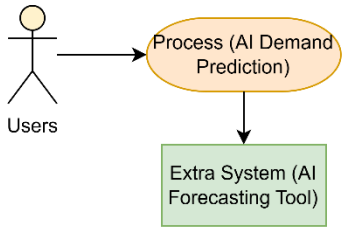
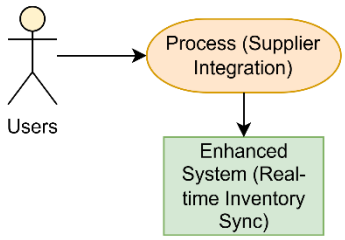
Option	Score
Solution 1:  Time: 8 - 12 months Cost: RM 2 - 3 million Advantages: enhances operational flow, improves efficiency Disadvantages: Requires integration with existing POS & suppliers Risks: System errors, supplier delays	Functionality: 5 Feasibility: 3 Alignment: 5 **Total Score: 13**
Solution 2:  Time: 6 - 9 months Cost: RM 1.5 - 2.5 million Advantages: Better inventory planning, prevents product expiration Disadvantages: requires historical data, ongoing updates required Risks: unexpected shortages or overstock	Functionality: 4 Feasibility: 4 Alignment: 4 **Total Score: 12**
Solution 3:  Time: 9 - 14 months Cost: RM 2 - 4 million Advantages: ensure smooth supply chain management, minimize manual ordering Disadvantages: complex implementation Risks: system failures	Functionality: 5 Feasibility: 2 Alignment: 5 **Total Score: 12**

Diagram 2.6.2 *Option Assessment: Efficient Inventory Distribution and Centralized Inventory*

Diagram 2.6.2 shows the option assessment of efficient inventory distribution and centralized inventory. This option assessment is to provide suggestions to address the issue of decentralized inventory in central region and southern region. The first solution is to implement a smart inventory system which includes automated restocking. The time required to implement this solution is 8 to 12 months with a cost of RM2 million to RM3 million. The advantages of this solution are it enhances the operational flow and efficiency of Buyomart. The disadvantage of this solution is it requires integration with the existing point of sales system and suppliers. The risks that might be faced are system errors and supplier delays.

The second solution is to implement the process of AI Demand Prediction and AI forecasting tool into the system. The time needed to implement this solution is 6 to 9 months with a cost of RM1.5 million to RM2.5 million. The advantages of this solution are the system can provide better inventory planning for managers and prevent large number of products from expiring. The disadvantages of this solution are it requires historical data, and ongoing updates are needed to maintain the system. The risk that will be faced are unexpected shortages or overstock.

The third solution is to integrate the real-time inventory synchronization with the current system. The time required to implement this solution is 9 to 14 months with an estimated cost of RM2 million to RM4 million. The advantages of this solution are ensuring a smooth supply chain management and minimize manual ordering. The disadvantage of this system is the complex implementation of the system. The system might also face a risk of system failures.

In this option assessment, solution 1, implement a smart inventory system has the highest total score of 13. Followed by both solution 2, implement the process of AI Demand Prediction and solution 3, integrate real-time inventory synchronization with a total score of 12.

Cloud-Based Sales Tracking

Options Assessments

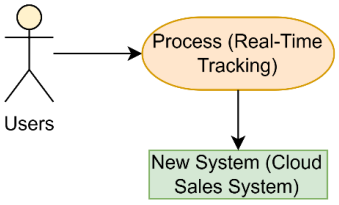
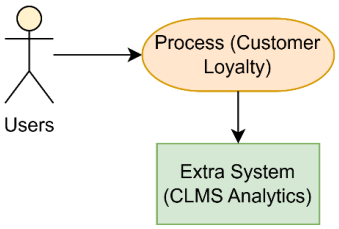
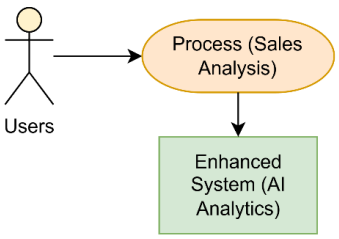
Option		Score
Solution 1: 	Time: 8 - 12 months Cost: RM 2 - 3 million Advantages: real-time visibility, reduces reporting time Disadvantages: high initial cost, requires staff training Risks: data security concerns, possible system downtimes	Functionality: 5 Feasibility: 3 Alignment: 5 **Total Score: 13**
Solution 2: 	Time: 6 - 10 months Cost: RM 1.5 - 2.5 million Advantages: personalized offers, improve customer retention Disadvantages: require app integration, additional costs for CLMS licensing Risks: customer data privacy risks, low adoption rates	Functionality: 4 Feasibility: 4 Alignment: 4 **Total Score: 12**
Solution 3: 	Time: 7 - 12 months Cost: RM 2 - 3 million Advantages: predict sales trends, optimize inventory management Disadvantages: requires data standardization, cloud storage expansion Risks: AI misinterpretation of trends, inaccurate forecasts	Functionality: 5 Feasibility: 3 Alignment: 4 **Total Score: 12**

Diagram 2.6.4 *Option Assessment: Cloud-Based Sales Tracking*

Diagram 2.6.4 shows the option assessment of cloud-based sales tracking. This option assessment provides recommendations for integrating a cloud-based sales tracking system to enhance real-time data accessibility. The first solution is to implement a real-time tracking process which often updates the stock levels of Buyomart branches to the cloud sales system. The time required to implement this solution is 8 to 12 months with a cost of RM2 million to RM3 million. This system provides real-time stock levels visibility for managers. It also reduces the reporting time because managers can access to stock-levels of different branches across different regions in real-time. The disadvantages of this solution are it requires high initial cost and training for the staffs to adopt to the latest system. The risks of this solution are data security concerns and possible system downtimes.

The second solution is to implement an extra system, which is to integrate the sales tracking system with a Customer Loyalty Management System (CLMS) to track customer loyalty points and redemption. The system would also provide personalized recommendations based on the customer's purchase history. The time required to implement this solution is 6 to 10 months with a cost of RM1.5 million to RM2.5 million. The advantages of this solution are providing personalized offers and improving the customer retention. The disadvantages of this system are it requires app integration and additional costs for CLMS licensing.

The third suggested solution is to implement sales analysis through utilizing AI analytics. The time required to implement this solution is 7 to 12 months with a cost of RM2 million to RM3 million. The advantages of this solution are to predict sales trends and optimize the inventory management of Buyomart. It can be a disadvantage as it requires data standardization between the branches and require cloud storage expansion. The risk of this solution is AI might misinterpret the stock trends which leads to inaccurate forecasts.

In this option assessment, solution 1, has the highest total score of 13, followed by solution 2 and solution 3 both with a total score of 12.

2.6.2 Solution Overviews

Solution Overviews

1. Overview and Goals

Overview

Buyomart is upgrading the Point-of-Sales (POS) system to enhance the transaction speed so that customers won't have to queue for a long time during payment. The current system has limited payment options and requires manual inventory updates which causes operational inefficiencies. Through implementing a modern POS system, Buyomart can enhance the customer experience and streamline transactions through real-time data synchronization.

Goals

- reduce checkout time and improve transaction speed
- enhance payment flexibility and security
- enhance staff productivity

2. Scope and Stakeholders

Scope

Buyomart will upgrade to a new point-of-sale (POS) system which improves the barcode scanning to reduce checkout time and improve transaction efficiency.

Stakeholders

Internal Stakeholders

- IT Team: Responsible in system integration, maintenance and troubleshooting
- Buyomart Management: Oversees project execution and investment
- Store Managers and Staff: end users of new POS and inventory system

External Stakeholders

- Customers: experience improved services
- Technology Vendors: Suppliers of POS systems

3. Essential Requirements

- require barcode scanners, updated POS terminals, receipt printers
- real-time transaction processing, multiple payment method integration
- encryption in payment processing, compliance with data protection regulations

4. Business Benefits

- speeds up checkout time
- minimize transaction errors
- improve payment processing
- secure transactions with encrypted payment

5. Capability impact

- Order Fulfillment (High)
- Order Management (Medium)
- Customer Analytics (Medium)

6. Involved Partners

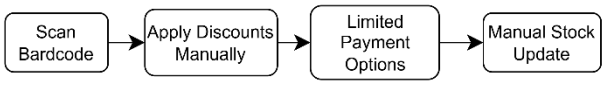
- StoreHub
- Visa, Mastercard, Paypal, GrabPay, or local e-wallet services

7. Estimations

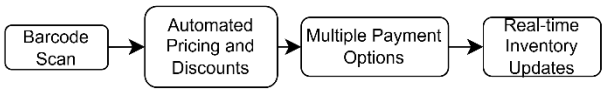
Time: 8 - 12 months
Cost: RM 2 - 3 million

8. Business Process Changes

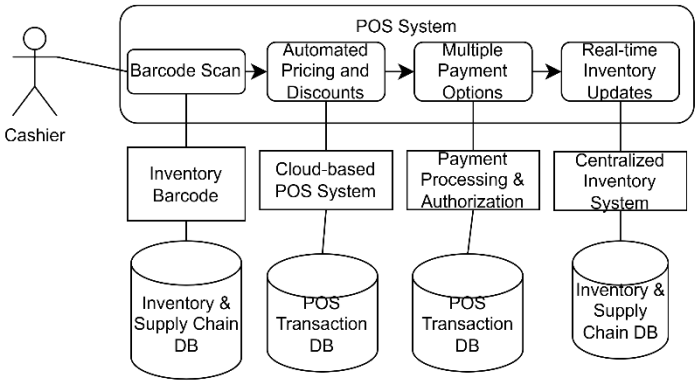
Before upgrade:



After upgrade:



9. Architectural Overview



10. Key Risks

- possible system integration issues

Diagram 2.6.5 *Solution Overview: Upgrading a Retail Point-of-Sales System & Checkout Process*

Diagram 2.6.5 shows the solution overview of upgrading the retail point of sales system and checkout process. It focuses on upgrading Buyomart's Point-of-Sales (POS)

system to enhance the transaction speed, customer's payment flexibility and the staff's productivity in carrying out the work. It also streamlines transactions through real-time data synchronization. This upgrade includes new barcode scanning machines, automated pricing and discounts, multiple payment options and real-time inventory updates. The stakeholders involved are IT Team, Buyomart management team, store managers, technology vendors and customers. The involved partners include StoreHub and local e-wallet services such as GrabPay, Paypal and so on. StoreHub is a platform which can digitise Buyomart's operations and automate business easily. This implementation is estimated to take 8 to 12 months with an estimated cost of RM2 million to RM3 million. Before the upgrade, Buyomart used outdated barcode scanners to scan the products. Cashiers had to manually apply the discounts and customers had limited payment options, either to pay by cash, debit cards or credit cards. The staffs had to manually count and update their stocks levels in Buyomart. After the upgrade, the cashier interacts with the system through scanning the barcodes of items, which triggers the automated pricing and discount calculations. Once the discount is applied to the total amount, customers can make their payment using multiple payment options. The system then processes and authorizes the payment, and the transaction details will be stored in the POS transaction database. Once payment is successful, the centralized inventory system updates the stock levels in the inventory and supply chain database.

Solution Overviews

1. Overview and Goals

Overview

In order to enhance Buyomart sales and delivery effectiveness, Buyomart will integrate the POS system with local delivery platforms like Foodpanda and GrabFood. This integration allow customers to place orders online and track deliveries in real-time. Buyomart improve customer satisfaction by automating order processing and delivery coordination.

Goals

- expand Buyomart's online ordering capabilities by integrating with third party delivery platforms
- enable real-time order tracking and multiple delivery options
- save time by automating order fulfillment between POS and delivery systems.

2. Scope and Stakeholders

Scope

Buyomart needs to integrate with local delivery platforms to improve digital sales and last-mile delivery services.

Stakeholders

Internal Stakeholders

- IT Team: Responsible in system integration, maintenance and troubleshooting
- Buyomart Management: Oversees project execution and investment

External Stakeholders

- Customers: Order and receive their items through delivery platforms
- Third party delivery services: collaboration for delivery integration

3. Essential Requirements

- assign delivery tasks to nearest available driver according to location
- able to track orders in real time
- verification steps for payments and delivery
- notifications to update on order status

4. Business Benefits

- increase sales and revenue as we reach to more customers
- reduces costs on the need for in-house delivery fleet
- increase competitive advantage

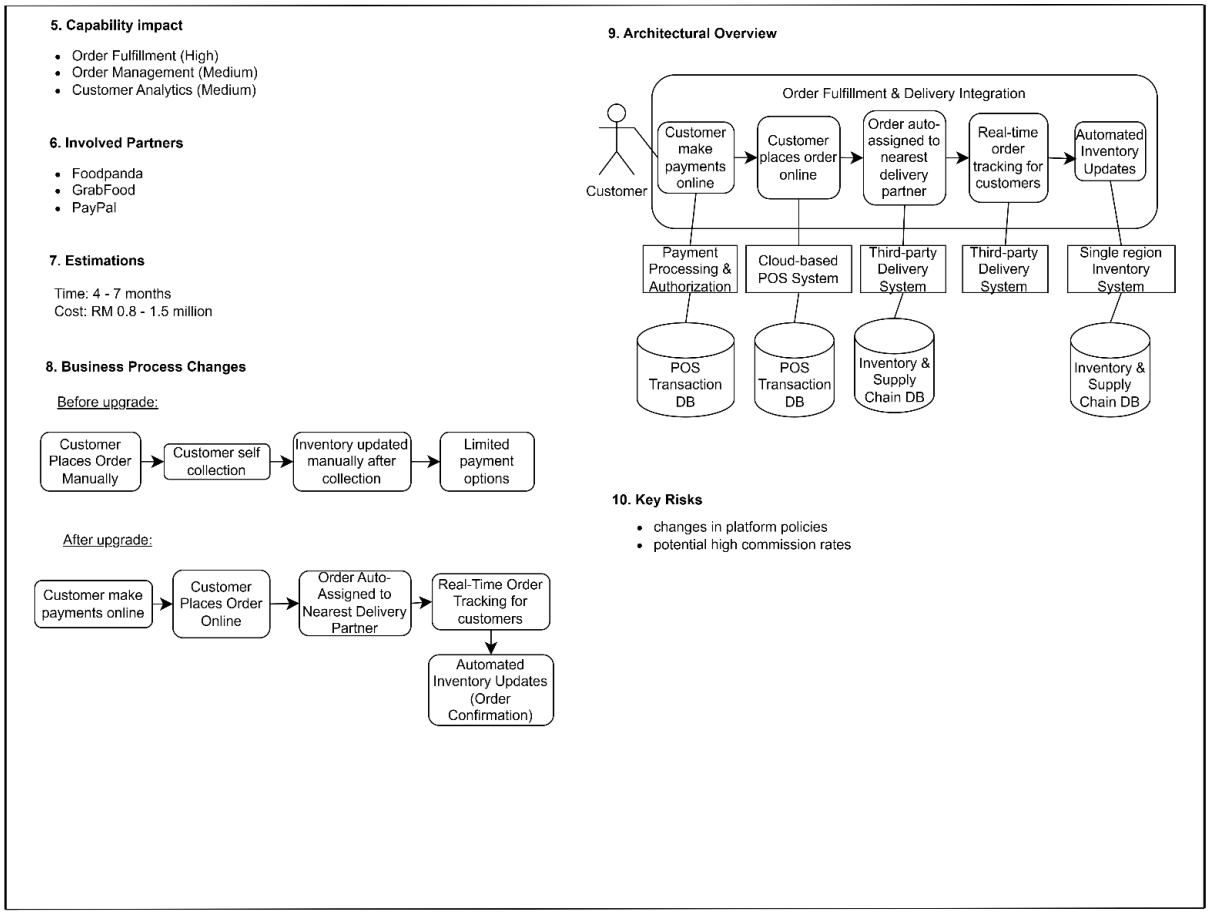


Diagram 2.6.6 *Solution Overview: Local Delivery App Collaboration*

Diagram 2.6.6 shows the solution overview of local delivery app collaboration. It focuses on enhancing Buyomart's sales by integrating with local delivery platforms, enabling real-time order tracking and automated order fulfilment between POS and delivery systems. This upgrade reduces Buyomart's costs on the need for in-house delivery fleet and increases its competitive advantage with other convenient stores. The upgrade includes real-time order tracking, customer notifications on order status, automated assignment of delivery tasks to nearest available driver and verification steps for payments. The stakeholders involved are IT team, Buyomart Management, customers and third-party delivery services. The involved partners are Foodpanda, GrabFood and PayPal. The estimated time to implement this solution is 4 to 7 months with a cost of RM0.8 million to RM1.5 million. Before the upgrade, customers had to place orders manually through calling the phone number of Buyomart stores. Once the order is ready, customers would self-collect their items in stores and the staff will manually update the store's inventory. Customers had limited payment options for making their payments in stores. After the upgrade, customers can make their payments online. The system will process and authorize their payments while storing transaction details in the POS Transaction database. Once payment is successful, the order will be placed and stored in the cloud-based system. Customer's order will we assigned to the nearest delivery partner through the third-party delivery system. Customers can track their orders in real-time via the third-party delivery system. Once they received their orders, the system will update the inventory and supply chain database.

Solution Overviews

1. Overview and Goals

Overview

Buyomart will implement a **centralized inventory management system** to streamline stock tracking and other data across all branches. The system helps to ensure real-time visibility of inventory levels and reduce risk of overstocking. Automated inventory restocking system will be integrated and demand trend analysis will be used to predict stock levels efficiently.

Goals

- minimize manual stock checks and streamline order management across multiple locations
- centralize stock tracking across all branches
- ensure products are always available, reduce delays

2. Scope and Stakeholders

Scope

Buyomart will include a centralized inventory management system and implement an automated inventory restocking system based on the demand trends for each branch.

Stakeholders

Internal Stakeholders

- IT Team: implement and maintain inventory management system and automation processes
- Inventory Management Team: analyze stock trends and manage replenishment

External Stakeholders

- Suppliers: Provide stock replenishment
- Logistics: handle transportation of goods to Buyomart's branches

3. Essential Requirements

- Tracks stock levels, detects low inventory
- AI-driven models to analyze sales trends and predict future inventory needs for each branch
- Ensures inventory data is updated across all branches
- Ensure inventory data is retrievable

4. Business Benefits

- Enhances operation flow
- Enhance supply chain efficiency by providing accurate demand forecasts
- Reduces operational costs by minimizing manual inventory checks

5. Capability impact

- Order Fulfillment (High)
- Order Management (High)
- Customer Analytics (Medium)

6. Involved Partners

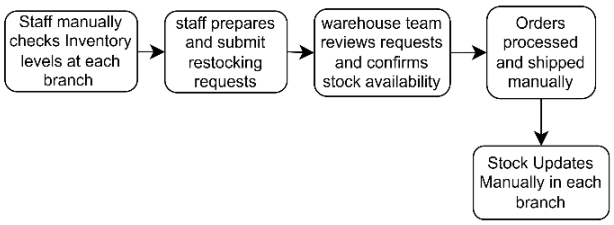
- IBM
- Microsoft Azure
- Tableau

7. Estimations

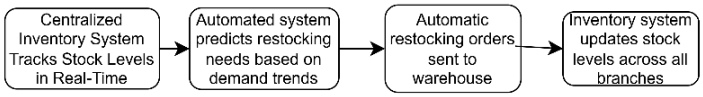
Time: 8 - 12 months
Cost: RM 2 - 3 million

8. Business Process Changes

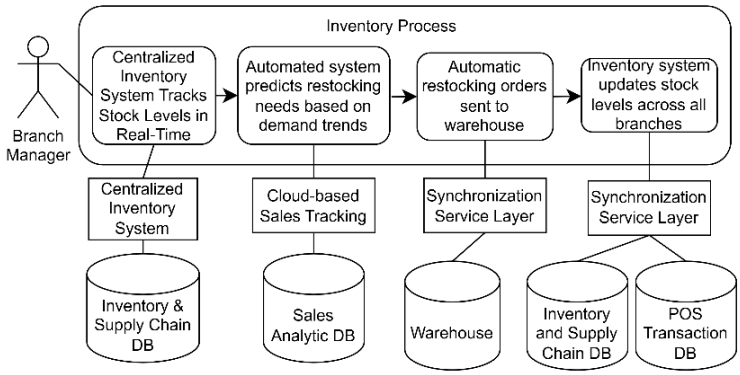
Before upgrade:



After upgrade:



9. Architectural Overview



10. Key Risks

- system errors
- supplier delays

Diagram 2.6.7 *Solution Overview: Efficient Inventory Distribution and Centralized Inventory*

Diagram 2.6.7 shows the solution overview of efficient inventory distribution and centralized inventory management. The system focuses on centralizing stock tracking and data across all branches. It helps managers to gain real-time visibility into inventory levels and predict the stock levels efficiently. It can also ensure that Buyomart's products remains constantly available. The system includes tracking stock levels, detecting low inventory, analyzing sales trends using AI-driven models, predicting future inventory needs and ensuring inventory data is updated across all branches. The stakeholders involved are IT Team, inventory management team, suppliers and logistics. The partners involved are IBM, Microsoft Azure and Tableau. The estimated time to implement this solution is 8 to 12 months with an estimated cost of RM2 million to RM3 million. Before the upgrade, the staff manually checked the inventory levels at each branch. Once the inventory check was completed, they prepared and submitted restocking requests. The warehouse team would then review these requests and confirm stock availability and manually process and ship orders. Once the stock was received, the staff had to manually update the inventory records in each branch. After the upgrade, the centralized inventory system will track stock levels in real-time, storing the data in the inventory and supply chain database. The automated system will predict restocking needs using the cloud-based sales tracking and the report will be stored in the sales analytic database. Automatic restocking orders will be sent to the warehouse through the synchronization service layer and the inventory system will update the stock levels across all branches.

Solution Overviews

1. Overview and Goals

Overview

In order to monitor and report sales across all branches, Buyomart aims to implement real-time cloud-based sales tracking system. The system will provide automated sales reporting analytics which helps to track revenue and sales trends in real time. Buyomart will ensure secure data storage, seamless access across branches and improved decision-making after implementing cloud technology.

Goals

- provide quick access to sales data across all branches
- optimize pricing, promotions and inventory planning
- store and protect sales data in cloud-based system

2. Scope and Stakeholders

Scope

Buyomart will implement a real-time cloud-based sales tracking system which provides automated sales reporting and analytics.

Stakeholders

Internal Stakeholders

- Inventory Managers: Utilize sales data to adjust inventory levels and restock schedules
- Senior Management and executives: Use analytics for strategic decision-making and performance evaluation
- IT Team: Manage system integration, security and technical maintenance of cloud-based solution

External Stakeholders

- Cloud Service Providers: host and maintain the cloud-based system
- Suppliers and distributors: adjust stock supply and distribution logistics based on sales data insights

3. Essential Requirements

- ensure that all branches sync their sales data to the central cloud system
- system which generates real-time reports
- data encryption, secure authentication and data protection regulations
- notifies users for unusual sales activity or low stock levels

4. Business Benefits

- enhances sales performance tracking
- help management make informed business decisions
- response faster towards market trends

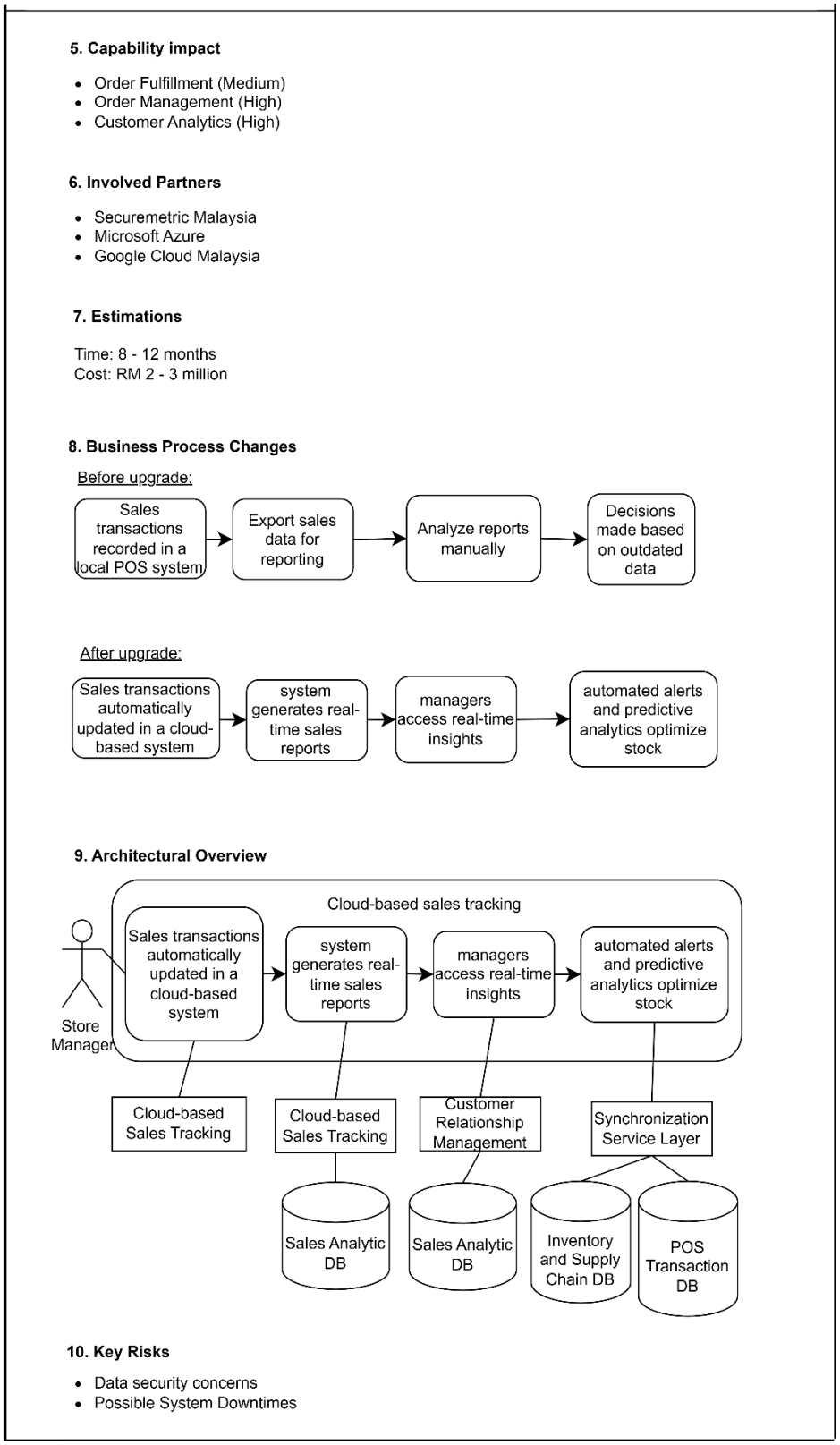


Diagram 2.6.8 *Solution Overview: Cloud-Based Sales Tracking*

Diagram 2.6.8 shows the solution overview of cloud-based sales tracking. The system focuses on implementing real-time cloud-based sales tracking system, where automated sales reporting will be implemented to monitor revenue and sales trends in real time. The upgrade aims to store and protect sales data in the cloud-based system. It also

provides managers quick access to sales data across all branches. This system will help optimize pricing, promotions and inventory planning. The solution includes synchronizing sales data with the central cloud system, generating real-time reports, notifying users for unusual sales activity or low stock levels and ensuring data encryption with secure authentication. The stakeholders involved in this solution include inventory managers, senior management, executives, IT team, cloud service providers, suppliers and distributors. The partners involved in this solution are Securemetric Malaysia, Microsoft Azure and Google Cloud Malaysia. The estimated time to implement this solution is 8 to 12 months with an estimated cost of RM2 million to RM3 million. Before the upgrade, sales transactions were recorded in the local POS system. Sales data has to be manually exported and compiled into reports by the staff. The managers required to analyze the reports and make decisions based on outdated data. After the upgrade, sales transactions will be automatically updated in a cloud-based system. The cloud-based sales tracking system will analyze the sales data and generate real-time sales reports, which will be stored in the sales analytic database. Managers will have real-time access to these reports through the customer relationship management system in the sales analytic database. The system will generate automated alerts and predictive analytics, optimizing stock levels by updating the inventory and supply chain database as well as the POS transaction database.

2.7 Designs

According to Kotusev (2016), designs are IT-focused Changes which provide detailed technical descriptions of specific IT projects actionable for project teams. The aim of implementing designs is to help implement IT projects according to business and architectural requirements. Architects, project teams and business representatives usually will collaborate to develop the design artifacts and project teams will use it to implement IT projects. The examples of EA artifacts related to Designs are solution designs, preliminary solution designs and so on.

2.7.1 Preliminary Solution Designs

Preliminary Solution Designs

1. Brief Overview

Buyomart will upgrade the Point of Sales (POS) system to improve transaction speed, staff productivity and secure transactions. The new system will feature automated discount applications, real-time inventory updates and various payment options.

2. Goals and Objectives

- enhance checkout speed to reduce waiting times
- real-time inventory management by automatically update stock levels
- improve staff productivity
- secure transactions

3. Scope and Stakeholders

Scope

- upgrade to new point of sales (POS) system
- improve barcode scanning to reduce checkout time
- software enhancement to modernize POS software

Stakeholders

Internal Stakeholders

- IT Team
- Buyomart Management
- Store Managers and Staff

External Stakeholders

- Customers
- Technology Vendors

4. Business Requirements

- integrate with multiple digital payment methods
- automated discount application and promotion
- implement real-time stock tracking
- secure transaction process

5. Involved Partners

- StoreHub
- Visa, Mastercard, Paypal, or local e-wallet services

6. Key Technologies

- Secure Payment Gateway Integration
- Cloud-Based POS Software
- Automated inventory management

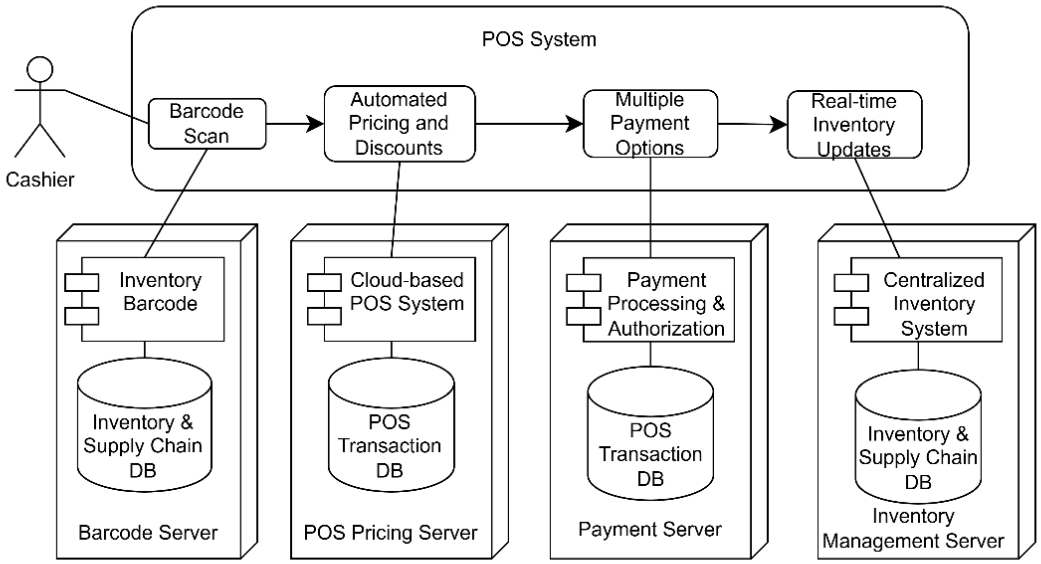
7. Accurate Estimations

Time: 8 - 12 months
Cost: RM 2 - 3 million

8.Technical Risks

- security risks in securing payment data and transaction details
- compatibility issues between new POS software and existing inventory
- incomplete backups which might cause data loss

9. High-Level Architecture



10. Application Interaction

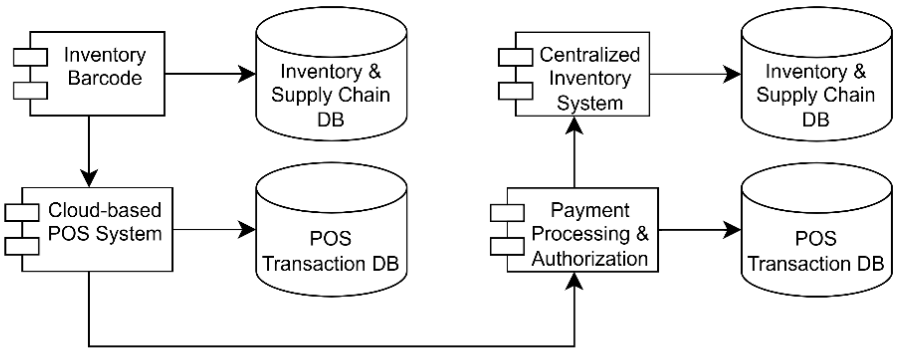


Diagram 2.7.0 *Preliminary Solution Design: Upgrading a Retail Point-of-Sales System & Checkout Process*

Diagram 2.7.0 shows the preliminary solution design of upgrading a retail point-of-sales system and checkout process. Buyomart plans to upgrade its point-of-sales system to enhance the transaction speed, improve staff productivity and secure transactions. The new system will integrate with multiple digital payment methods, automate discount application and promotion, implement real-time stock tracking and secure the transaction process. The scope of this solution is to implement the new point of sales system (POS), improve barcode scanning to reduce checkout time and enhance the software to modernize the POS software. The key technologies that will be used are secure payment gateway, cloud-based POS software and automated inventory management. One of the technical risks of this solution is the security risk in securing payment data and transaction details. Buyomart might also face compatibility issues between new POS software and the existing inventory as well as incomplete backups which might cause data loss. When the cashier scans the items, it will trigger the barcode server, which retrieves the inventory details from the Inventory & Supply Chain Database. The Cloud-based POS system processes the scanned items and apply any discounts or promotions. The data is recorded in the POS Pricing Server and stored in the POS Transaction Database. Once the total amount is finalised, customers can choose from various payment methods. It triggers the payment server, where Payment Processing and Authorization application processes the payment and stores the transaction details into the POS Transaction database. Once the payment is successful, the centralized inventory system will update the stock levels in the inventory and supply chain database.

Preliminary Solution Designs

1. Brief Overview

Buyomart integrate the POS system with local delivery platforms like Foodpanda and GrabFood to enable real-time tracking, automate order processing and delivery coordination.

2. Goals and Objectives

- automate order fulfillment between POS and delivery systems
- enabling direct delivery from Buyomart Stores
- Real-time delivery tracking for customers

3. Scope and Stakeholders

Scope

- integrate with local delivery platforms to improve digital sales
- include real-time order tracking

Stakeholders

Internal Stakeholders

- IT Team
- Buyomart Management

External Stakeholders

- Customers
- Third party delivery services

4. Business Requirements

- Include verification steps for payments
- Tracking orders in real-time function should be included
- Notifications to update on order status
- Assign delivery tasks to nearest available driver

5. Involved Partners

- Foodpanda
- GrabFood
- PayPal

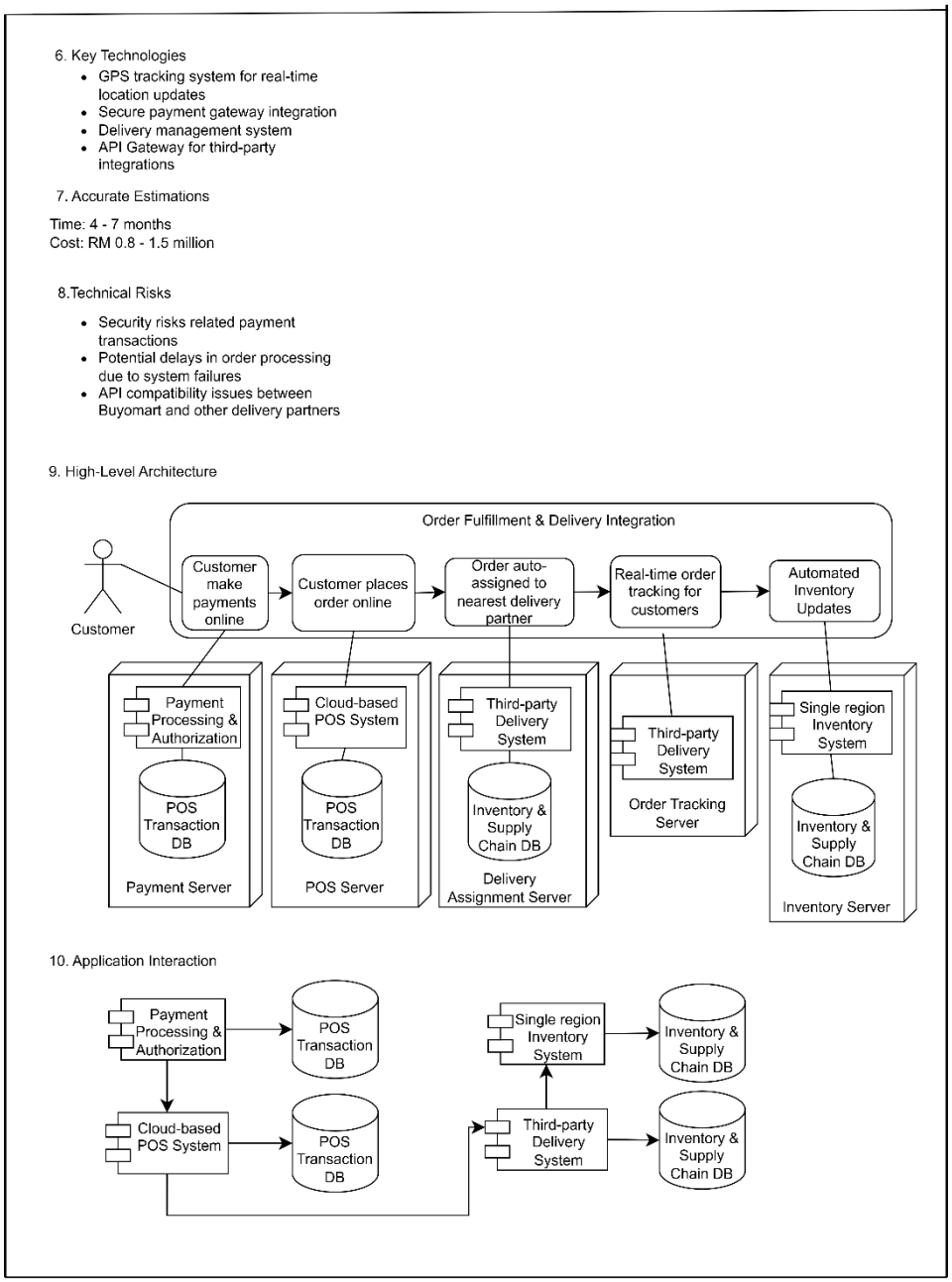


Diagram 2.7.1 *Preliminary Solution Design: Local Delivery App Collaboration*

Diagram 2.7.1 shows the preliminary solution design of the local delivery app collaboration. Buyomart plans to integrate the POS system with the local delivery platforms to enable real-time delivery tracking as well as automate order processing and delivery coordination. The new system will include verification steps for payments, track orders in real-time, notify customers on the order status and assign delivery tasks to nearest available driver. The key technologies that will be used are GPS tracking system, secure payment gateway, delivery management system and API gateway. The technical risks of this solution are security risks related to payment transactions, potential delays in order processing and API compatibility issues between Buyomart and other delivery partners. In this solution, when the customer makes payments online, the payment processing and authorization application will process the payments and store the transaction details into the POS Transaction database. Once the payment is successful, the cloud-based POS system in the

POS server will store the order details into the POS transaction database. The third-party delivery system will auto-assign a delivery partner for the order, and the driver's details will be stored into the Inventory and Supply Chain database. Customers can track their order in real time through the order tracking server. Once the order is delivered, the single region inventory system in the inventory server will automatically update the order status into the Inventory and Supply Chain database.

Preliminary Solution Designs

1. Brief Overview

Buyomart plans to implement a centralized inventory management system with an automated restocking system to enhance the inventory efficiency. The system will be used to analyze demand patterns and predict stock levels which can avoid the problems of stockouts and overstocking.

2. Goals and Objectives

- centralize stock tracking across all branches
- automate stock replenishment
- improve supply chain efficiency
- minimize manual stock checks

3. Scope and Stakeholders

Scope

- implement automated inventory restocking system
- include centralized inventory management system
- include real-time visibility of inventory levels function

Stakeholders

Internal Stakeholders

- IT Team
- Inventory Management Team

External Stakeholders

- Suppliers
- Logistics

4. Business Requirements

- AI-driven models to analyze sales patterns and predict future inventory needs
- Track stock levels, detects low inventory
- Ensure inventory data is updated across all branches

5. Involved Partners

- IBM
- Microsoft Azure
- Tableau

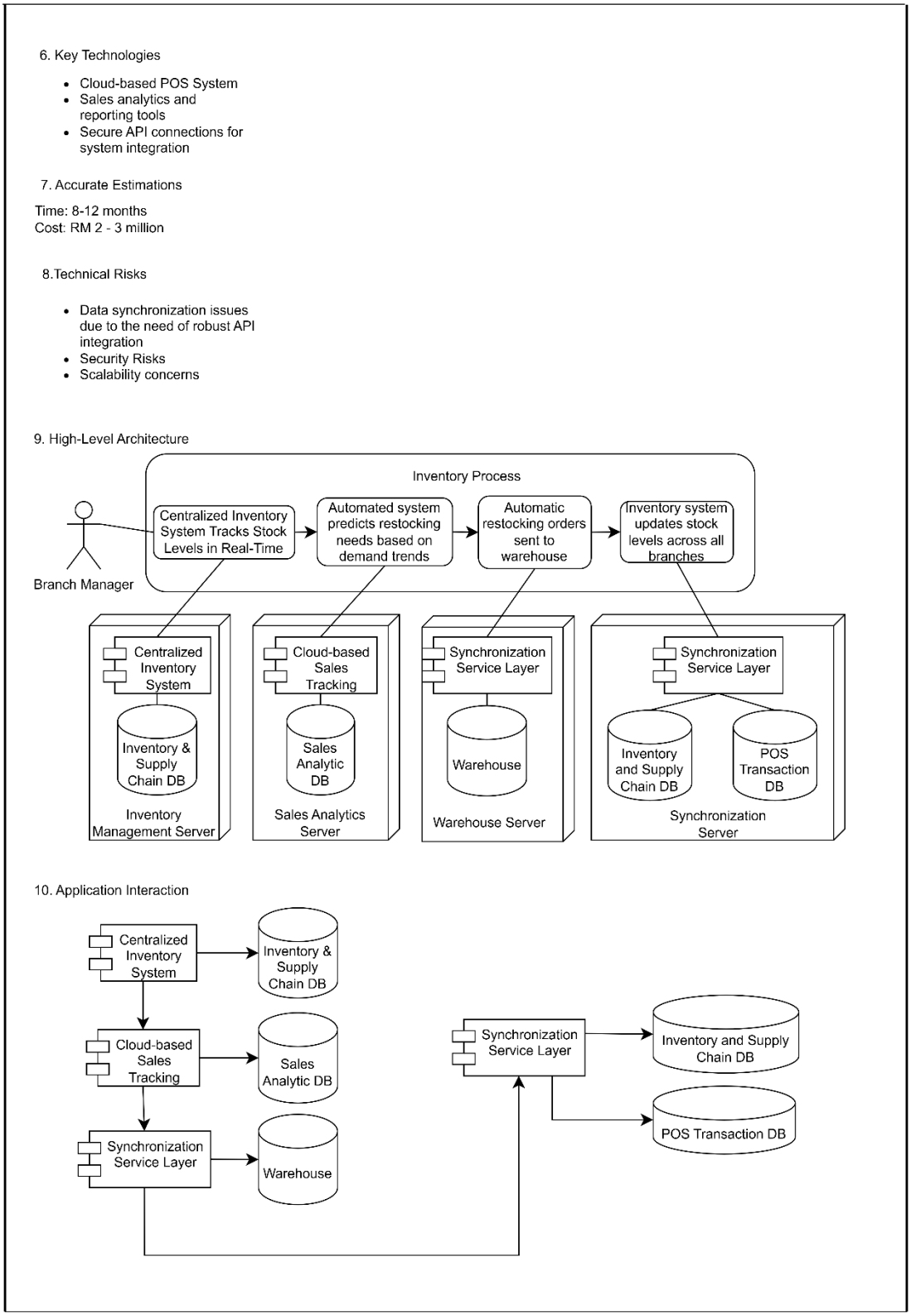


Diagram 2.7.2 *Preliminary Solution Design: Efficient Inventory Distribution and Centralized Inventory*

Diagram 2.7.2 shows the preliminary solution design of efficient inventory distribution and centralized inventory. Buyomart plans to implement a centralized inventory management system with an automated restocking system to improve the inventory efficiency and analyze the demand patterns to predict stock levels. Buyomart aims to

centralize stock tracking across all branches, automate stock replenishment, improve supply chain efficiency and minimize manual stock checks. The new system will incorporate AI-driven models to analyze sales patterns and predict future inventory needs. It will track stock levels, detect low inventory and ensure that the data is updated across all branches in real-time. The key technologies include cloud-based POS system, sales analytics and reporting tools as well as secure API connections for system integration. The technical risks include data synchronization issues, security risks and scalability concerns. In this solution, the branch manager can access the centralized inventory system which tracks stock levels in real-time. The Inventory Management server retrieves sales data from the Inventory and Supply Chain database. The Cloud-based Sales Tracking Application in Sales Analytics Server utilizes the retrieved data to predict restocking needs, the generated report is stored in the Sales Analytics database. Once the system completes the prediction, it automatically sends restocking orders to the warehouse server through the Synchronization Service Layer. The inventory system then updates the stock levels across all branches through the Synchronization Server.

Preliminary Solution Designs

1. Brief Overview

Buyomart implements a real-time cloud-based sales tracking system to monitor and report sales across all branches. The systems allows managers access to sales data, improve decision-making of stakeholders and automate reporting.

2. Goals and Objectives

- provides real-time monitoring of sales transactions across all branches
- automate sales reporting and analytics to reduce manual work
- provide insights based on sales trends
- enhance data security and accessibility

3. Scope and Stakeholders

Scope

- secure storage of sales data
- implement automated analytics and reporting tools
- Integrate the POS system with cloud-based sales tracking

Stakeholders

Internal Stakeholders

- Inventory Managers
- IT Team
- Senior Management and executives

External Stakeholders

- Cloud Service Providers
- Suppliers and distributors

4. Business Requirements

- A real time cloud-based sales database
- Integration with existing POS and inventory management systems
- Analytics dashboard for sales performance tracking.

5. Involved Partners

- Microsoft Azure
- Securemetric Malaysia
- Google Cloud Malaysia

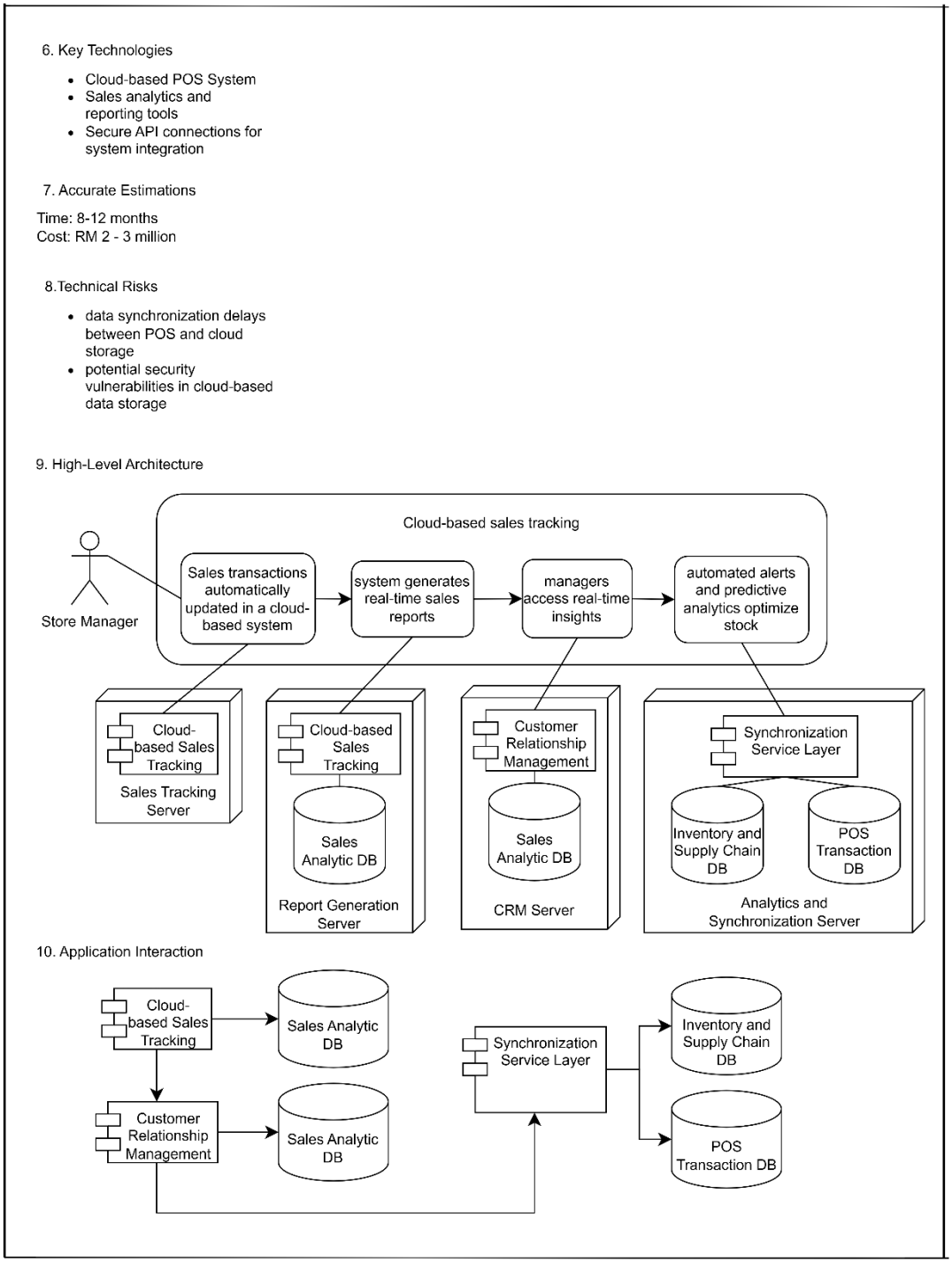


Diagram 2.7.3 *Preliminary Solution Design: Cloud-Based Sales Tracking*

Diagram 2.7.3 illustrates the preliminary solution design of cloud-based sales tracking. Buyomart plans to implement a real-time cloud-based sales tracking system to monitor and report sales across all branches. The system aims to provide automated sales reporting and analytics, generate insights based on sales trends as well as enhance data security and accessibility. The system will secure storage of sales data, implement automated analytics and reporting tools and integrate the POS system with cloud-based sales tracking. The system includes real-time cloud-based sales database, integration with existing POS and inventory management systems and analytics dashboard for sales performance tracking. The key technologies include cloud-based POS system, sales analytics and reporting tools

as well as secure API connections for system integration. The potential technical risks include data synchronization delays between POS and cloud storage, and potential security vulnerabilities in cloud-based data storage. In this solution, the store manager can access to the cloud-based sales tracking system. Sales transactions will be automatically updated in a cloud-based system. The system generates real-time sales reports through the Report Generation server, where the cloud-based sales tracking application will store them in the Sales Analytic database. Managers can access real-time insights through the CRM server where the customer relationship management application retrieves reports through the Sales Analytic database. The analytics and synchronization server automatically predicts trends and generates alerts for the Inventory and Supply Chain database and POS transaction database to optimize stock levels.

2.7.2 Solution Designs

Solution Designs

1. Brief Overview

Buyomart plans to upgrade the Point-of-Sales (POS) system to enhance the transaction speed so that customers won't have to queue for a long time during payment. The current system has limited payment options and requires manual inventory updates which causes operational inefficiencies. Through implementing a modern POS system, Buyomart can enhance the customer experience and streamline transactions through real-time data synchronization.

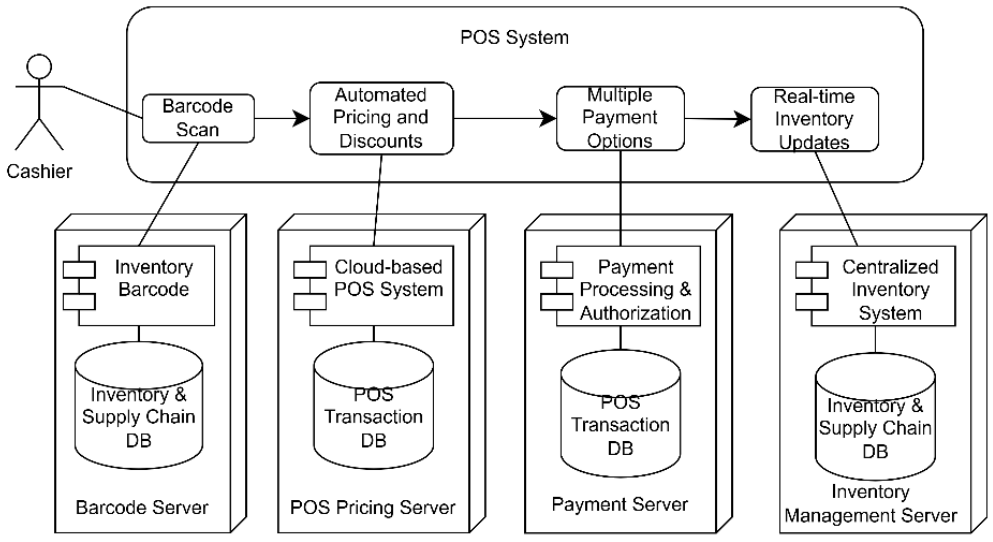
2. Goals and Objectives

- implement real-time inventory updates
- enhance payment flexibility and security
- enhance staff productivity
- automate pricing and discounts

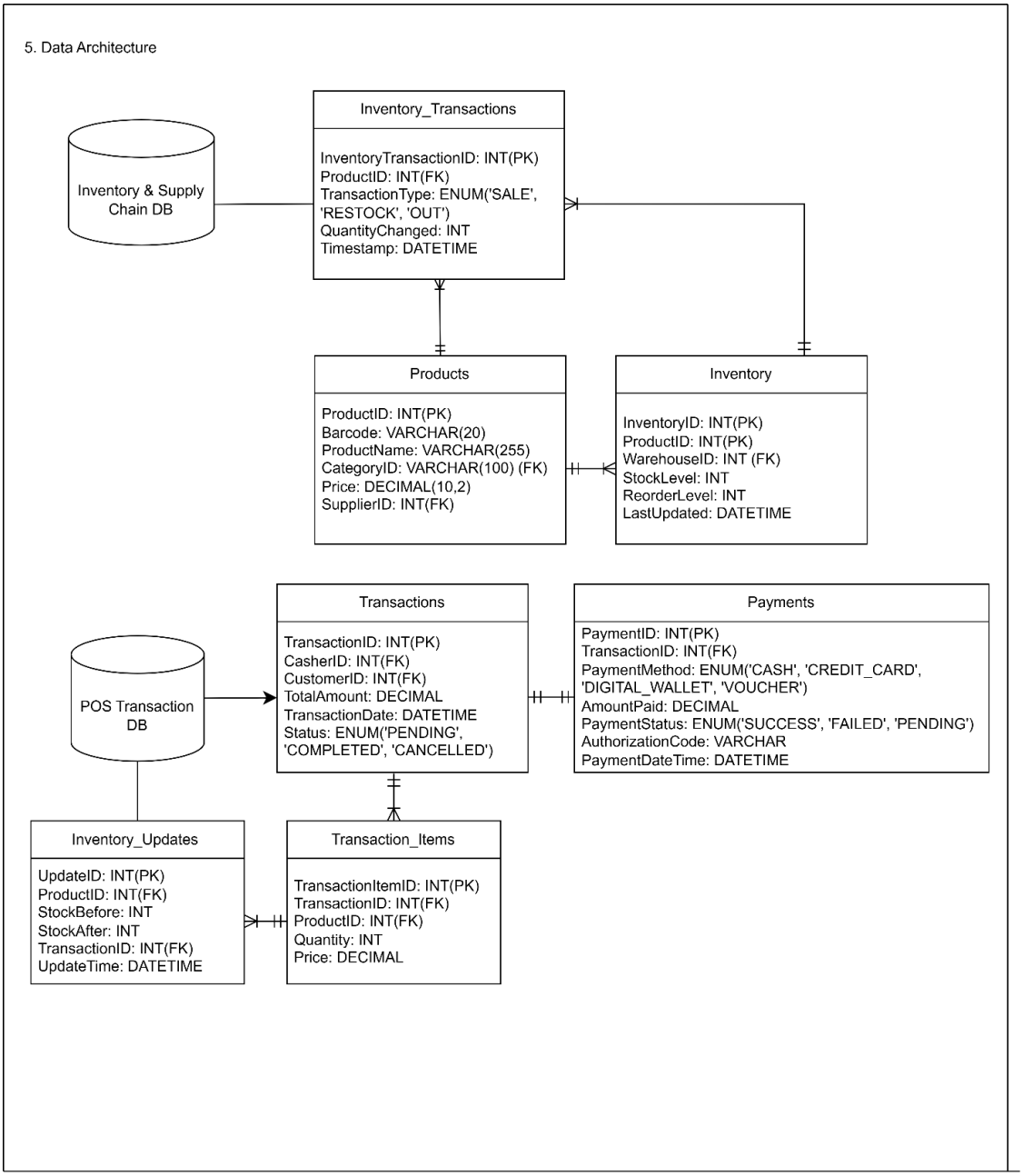
3. Detailed Requirements

- support barcode scanners, updated POS terminals, and receipt printers for seamless transaction processing
- provide real-time inventory updates after each transaction
- ensure seamless integration with third-party payment gateways
- implement end-to-end encryption for all payment transactions
- handle increased transaction load during peak hours
- implement automatic backup system to prevent data loss

4. Solution Context



5. Data Architecture



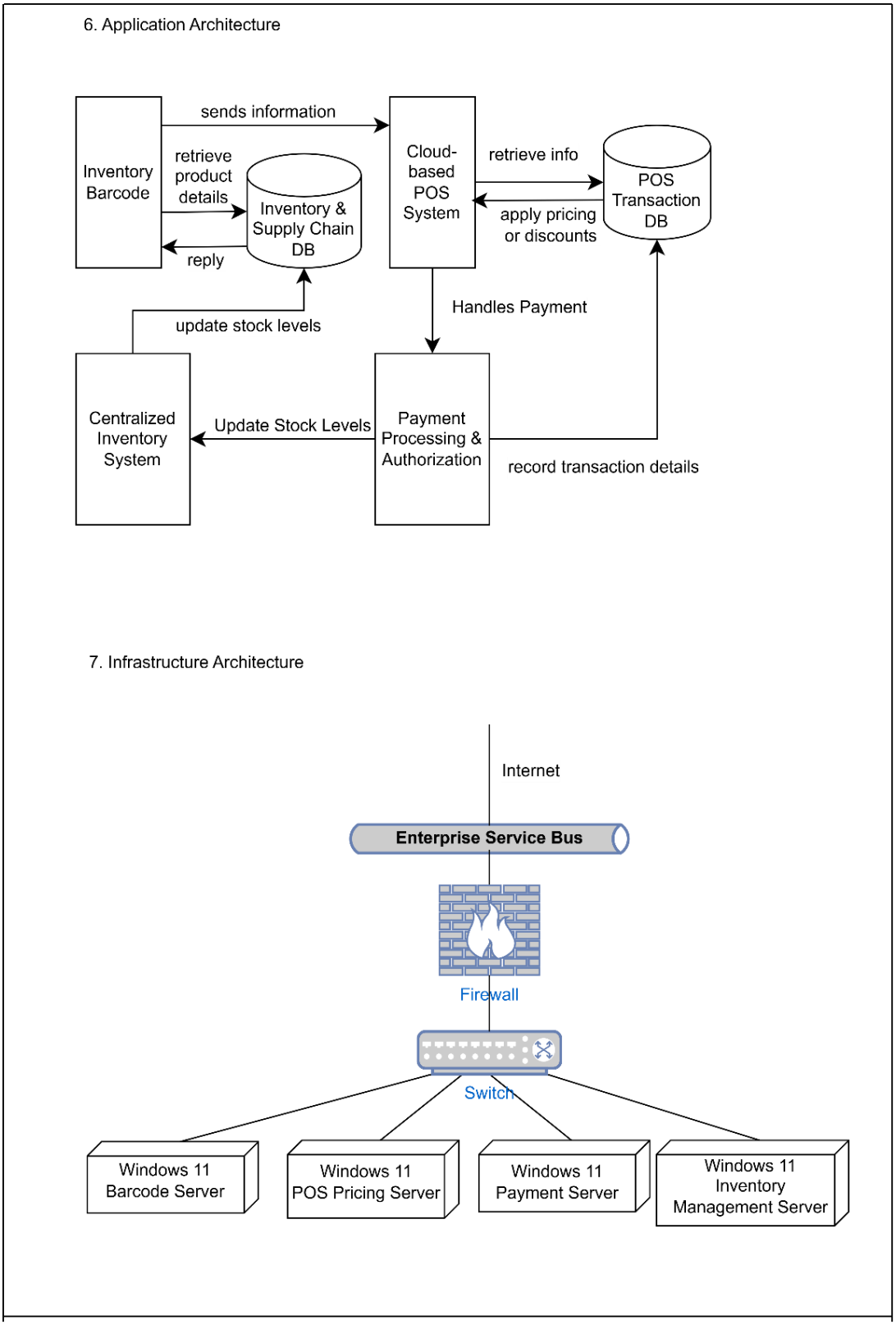


Diagram 2.7.4 *Solution Design: Upgrading a Retail Point-of-Sales System & Checkout Process*

Diagram 2.7.4 shows the solution design of upgrading a retail point-of-sales system and checkout process. Buyomart upgrades the point-of-sales (POS) system through implementing real-time inventory updates, enhancing payment flexibility and security as well

as automating pricing and discounts. One of the detailed requirements of this solution is that the system should support barcode scanners, updated POS terminals and receipt printers for seamless transaction processing. The system should be able to provide real time inventory updates after each transaction, ensure seamless integration with third-party payment gateways, implement end-to-end encryption for all payment transactions, handle increased transaction load during peak hours and implement automatic backup system to prevent data loss. The databases involved in the POS system are Inventory and Supply Chain database and POS Transaction database. The Inventory and Supply Chain database consists of tables named Inventory_Transactions, Products, Inventory whereas the POS Transaction database consists of tables named Transactions, Payments, Inventory_Updates and Transaction_Items.

In terms of application architecture, when the staff scans the item barcode, the inventory barcode retrieves the product details from the Inventory and Supply Chain database and sends them to the cloud-based POS system. The POS system then retrieves pricing and discounts information from the POS Transaction database and applies it to the total amount calculation. The payment processing and authorization application processes the customer's payment and records the transaction details in the POS Transaction database. Once payment is successful, the centralized inventory system updates stock levels in the Inventory and Supply Chain database. For the infrastructure architecture, the four key servers, Barcode server, POS Pricing Server, Payment Server and Inventory Management Server are connected to a switch, ensuring data transmission and fast communication between different components of the POS system. The switch is connected to the firewall, which secures the network by filtering the traffic between the internal servers and the Enterprise Service Bus.

Solution Designs

1. Brief Overview

Buyomart integrates local delivery platforms with the POS system such as Foodpanda and GrabFood, which allows real-time tracking, automated order processing and delivery coordination.

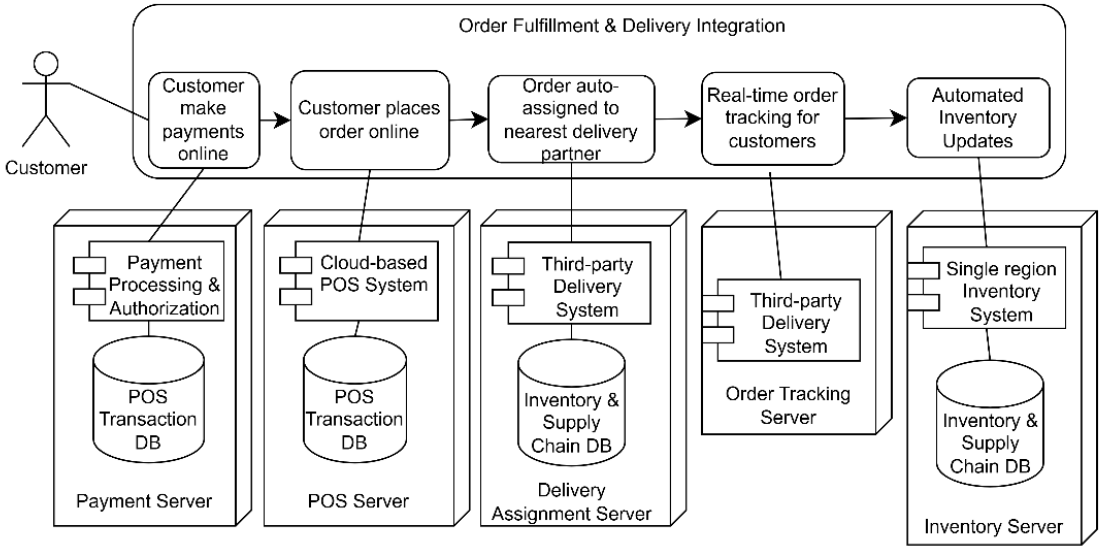
2. Goals and Objectives

- enables real-time delivery tracking for customers
- provides direct delivery service from Buyomart Stores
- automate order fulfillment between point of sales (POS) system with local delivery platforms

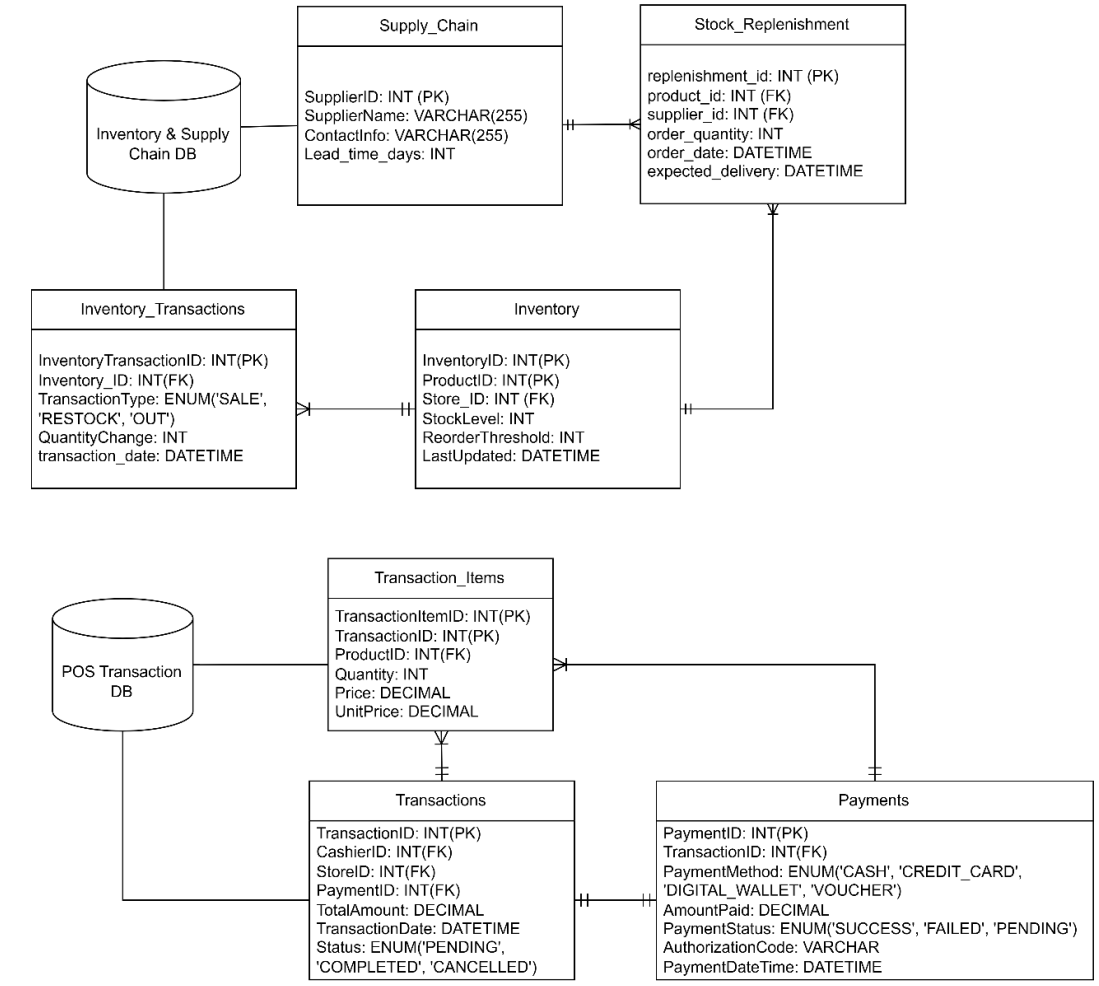
3. Detailed Requirements

- assign orders to the nearest available driver
- reassign the task to another driver if the previous driver is delayed
- validate payment details before confirming a transaction
- automatic order updates through notifications
- track orders in real-time function should be included

4. Solution Context



5. Data Architecture



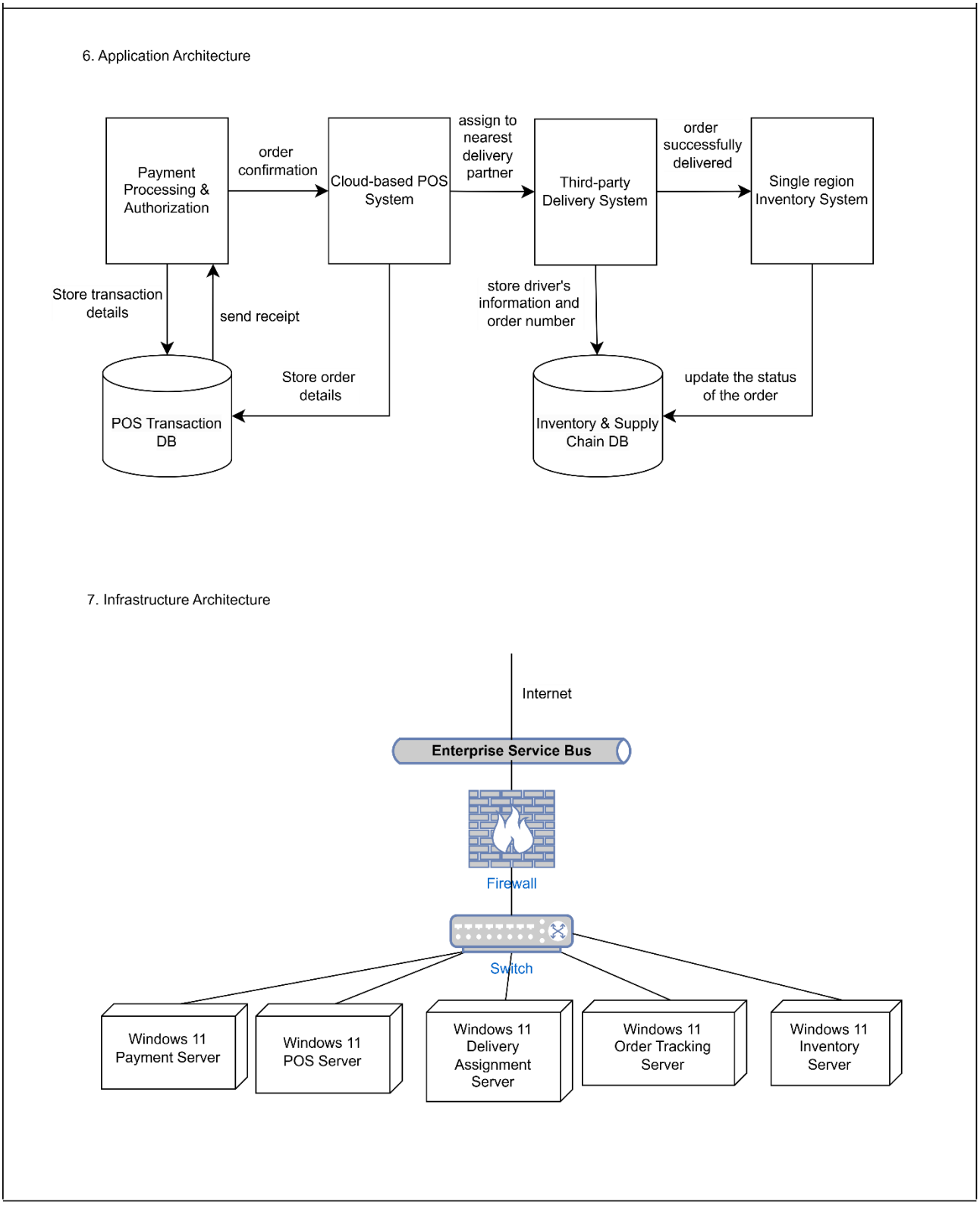


Diagram 2.7.5 *Solution Design: Local Delivery App Collaboration*

Diagram 2.7.5 shows the solution design for local delivery app collaboration. Buyomart integrates local delivery platforms with its point-of-sales system to enable real-time order tracking for customers, provide direct delivery services from Buyomart Stores and automate order fulfilment between point-of-sales system with local delivery platforms. One of the detailed requirements is to assign orders to the nearest available driver and automatically assign the task to another driver if the previous driver is delayed. The system

should be able to validate payment details before confirming a transaction, notify users on order updates and track orders in real-time. The databases involved in the Order Fulfilment and Delivery Integration system includes Inventory and Supply Chain database and POS transaction database. The Inventory and Supply Chain database consists of tables such as Supply_Chain, Stock_Replenishment, Inventory and Inventory_Transactions. The POS Transaction database consists of tables like Transaction_Items, Transactions and Payments.

In terms of application architecture, when the customers make a payment, the payment processing and authorization application processes the transaction and stores details in the POS Transaction database. Once the payment is successful, an order confirmation is sent to the cloud-based POS system, which then stores the order details in the POS Transaction database. The third-party system assigns the order to the nearest available delivery partner and store's the driver's information along with the order number in the Inventory and Supply Chain database. Once the order is successfully delivered, the single region inventory system will update the order status into the Inventory and Supply Chain database. In the infrastructure architecture, the Payment Server, POS Server, Delivery Assignment Server, Order Tracking Server and the Inventory Server are connected to a switch. The switch is connected to a firewall which secures the network through filtering the traffic between the internal servers and Enterprise Service Bus.

Solution Designs

1. Brief Overview

Buyomart implements a centralized inventory management system with an automated restocking system to improve inventory efficiency. The system can analyze demand patterns and estimate stock levels to avoid stockouts and overstocking problems.

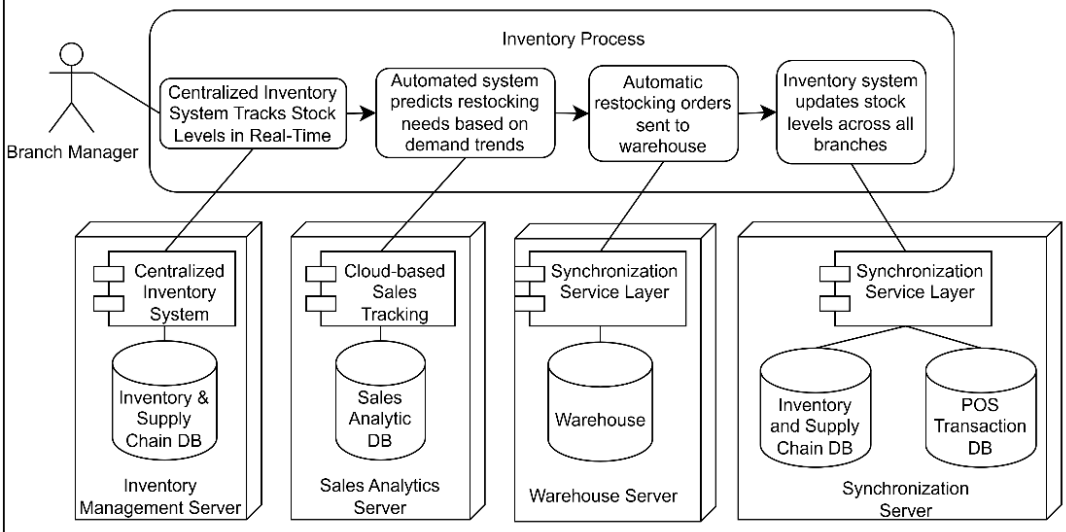
2. Goals and Objectives

- minimize manual stock checkings
- improve supply chain efficiency
- automate stock replenishment
- centralize stock tracking across all branches
- utilize AI-driven models to analyze sales patterns

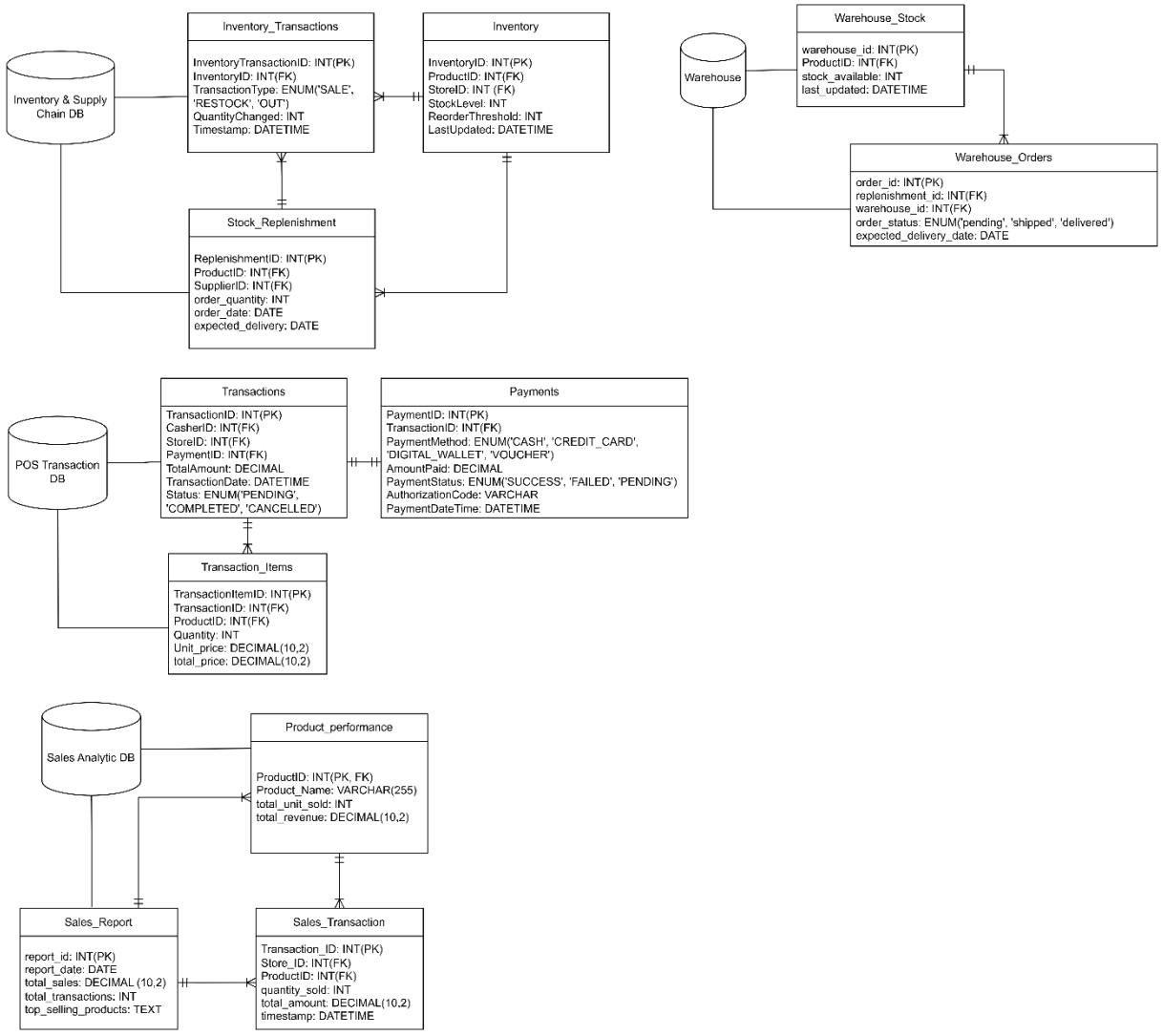
3. Detailed Requirements

- analyze past sales data and identify top-selling products
- suggest optimal stock levels, notify potential stock shortages
- track the latest stock levels and often update the centralized database
- secure customer and sales data with encryption

4. Solution Context



5. Data Architecture



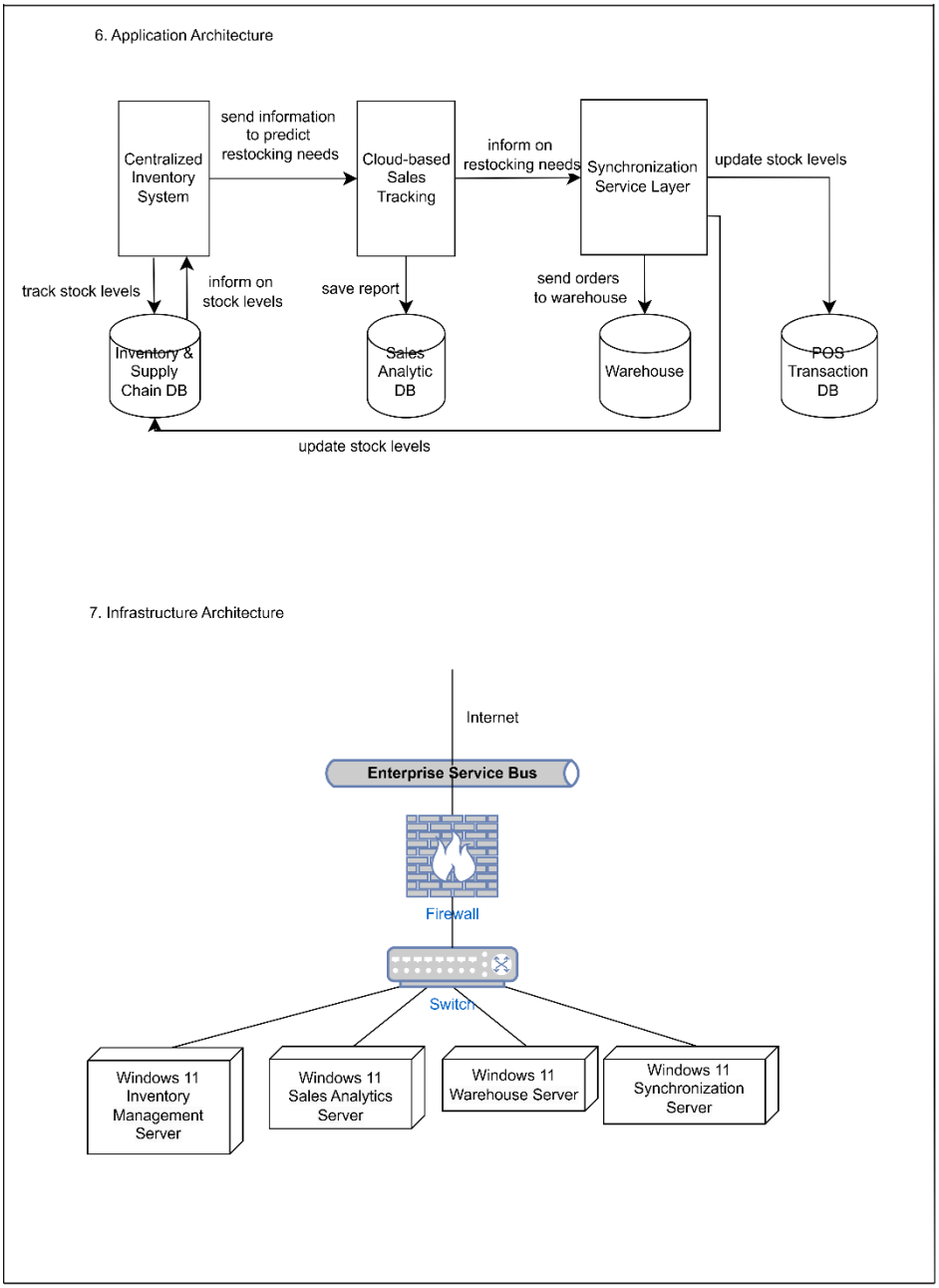


Diagram 2.7.6 *Solution Design: Efficient Inventory Distribution and Centralized Inventory*

Diagram 2.7.6 illustrates the solution design of efficient inventory distribution and centralized inventory. To improve inventory efficiency, Buyomart plans to implement a centralized inventory management system with an automated restocking system capable of analyzing demand patterns and estimating stock levels. The goal of this system is to minimize manual stock checking, improve supply chain efficiency, automate stock replenishment, centralize stock tracking across all branches and utilize AI-driven models to analyze sales patterns. The system includes features such as analyzing past sales data, identifying top-selling products, suggesting optimal stock levels which notify potential stock shortages, tracking the latest stock levels, updating the centralized database and securing customer and sales data with encryption. The databases included in the inventory process are Inventory and Supply Chain database, Sales Analytic database, Warehouse database and POS Transaction database. The Inventory and Supply Chain database includes tables

like Inventory_Transactions, Inventory and Stock_Replenishment whereas the Warehouse database includes tables like Warehouse_Stock and Warehouse_Orders. Besides, the POS Transaction database consists of tables like Transactions, Payments and Transaction_Items, whereas the Sales Analytic database includes tables like Sales_Report, Product_performance and Sales_Transaction.

In terms of Application Architecture, the centralized inventory system regularly tracks and updates stock levels in the Inventory and Supply Chain database. The cloud-based sales tracking receives this information sent by the centralized inventory system to predict restocking needs. Once the prediction report is generated, it is stored in the Sales Analytic database. The cloud-based sales tracking informs the restocking needs to the synchronization service layer, it updates stock levels in both the POS Transaction database and Inventory and Supply Chain database. The synchronization service layer sends restocking orders to warehouse for stock replenishment. In the infrastructure architecture, the servers, Inventory Management Server, Sales Analytics Server, Warehouse Server and Synchronization Server are connected to a switch. A firewall is placed between the swtich and the Enterprise Service Bus to filter traffic and enhance security.

Solution Designs

1. Brief Overview

Buyomart plans to include a real-time cloud-based sales tracking system to monitor and report sales across all the branches. The enhanced system allows managers to access to sales data, improve decision-making of stakeholders and automate reporting.

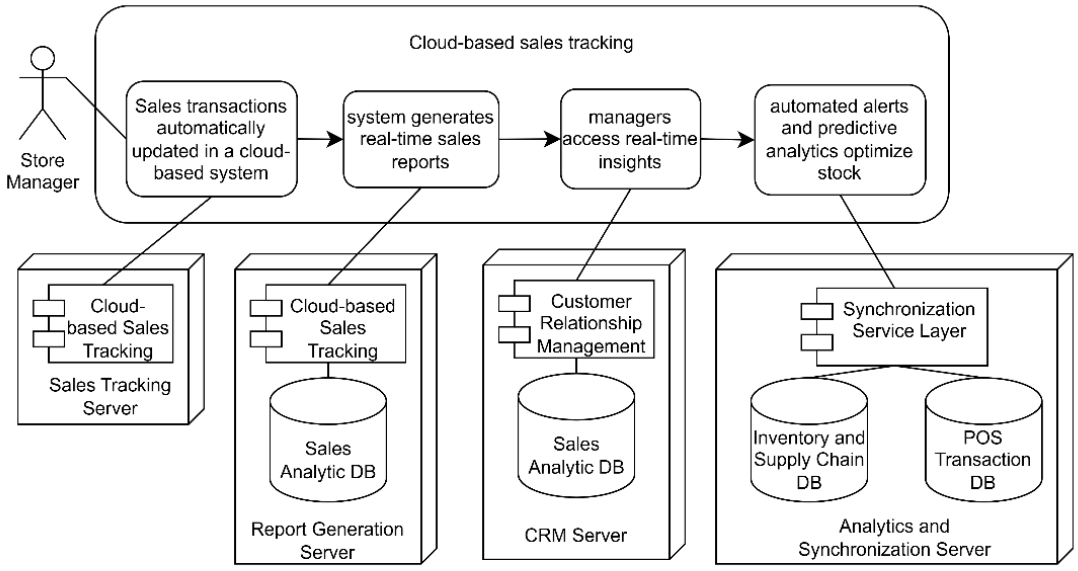
2. Goals and Objectives

- improve data security and accessibility
- show insights about the sales trends
- automate sales reporting and analytics to reduce manual work
- implement real-time monitoring of sales transactions across all branches

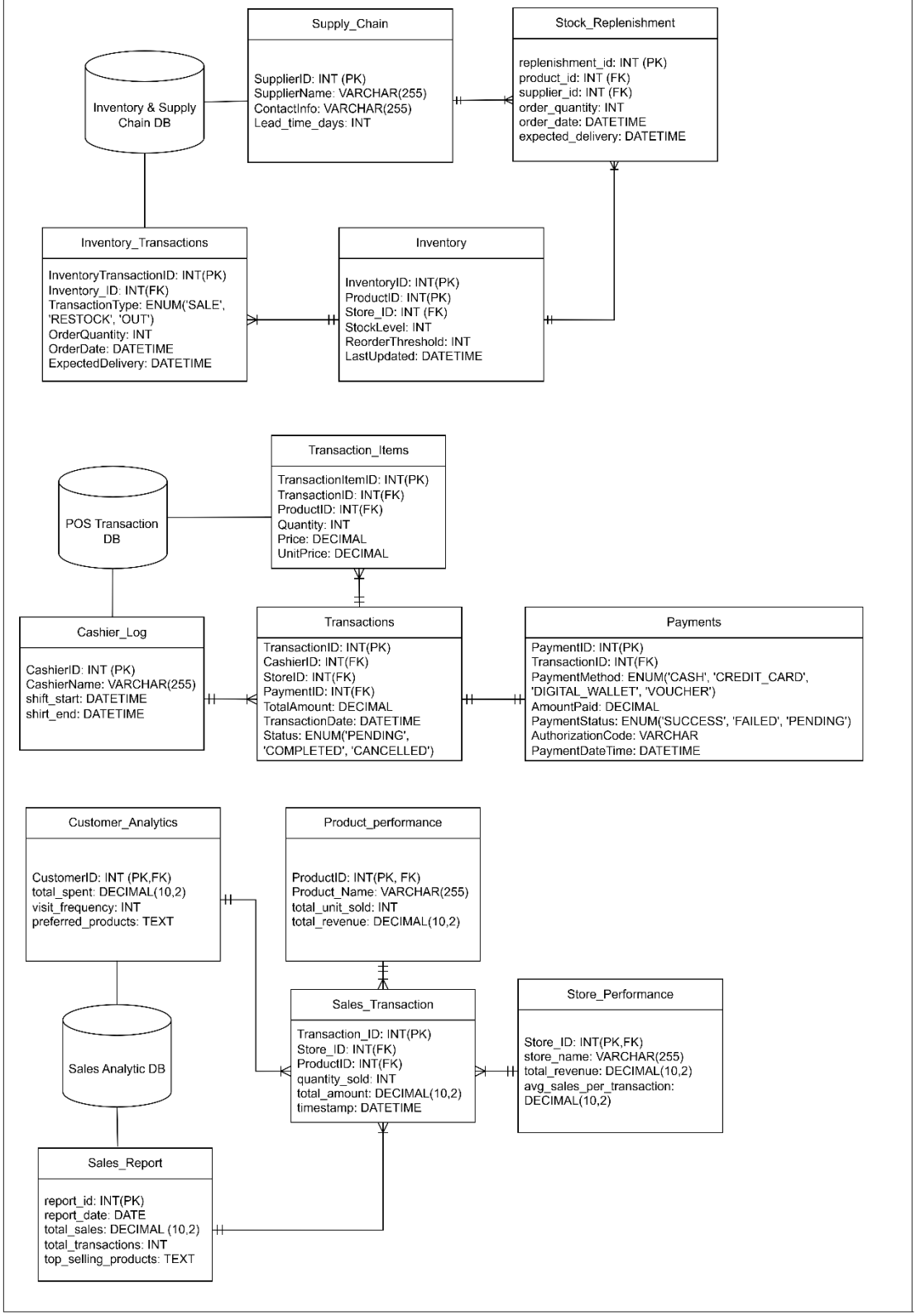
3. Detailed Requirements

- store real-time sales transactions (cloud-based) in the database
- able to handle network disruptions
- security measures that restrict unauthorized modifications
- perform daily and hourly backups
- integrate with existing POS and inventory management systems
- shows analytics dashboard on sales performance

4. Solution Context



5. Data Architecture



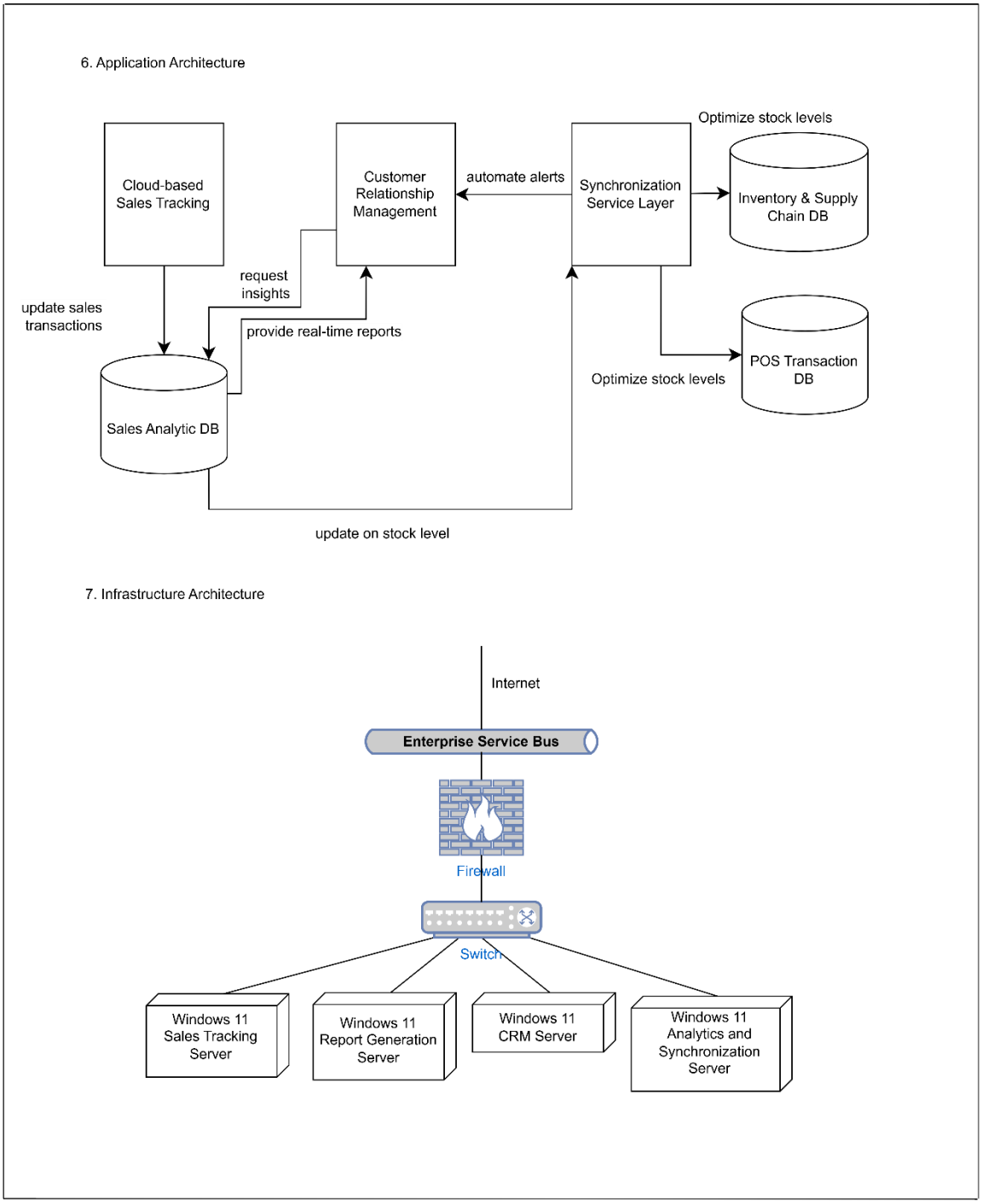


Diagram 2.7.7 *Solution Design: Cloud-Based Sales Tracking*

Diagram 2.7.7 shows the solution design of cloud-based sales tracking. Buyomart plans to implement a real-time cloud-based sales tracking system to monitor and report sales across all their branches. This system aims to improve data security and accessibility, show insights on the sales trends, automate sales reporting and analytics and implement real-time monitoring of sales transactions. The system includes storing real-time sales transactions in the database, handling network disruptions, restricting unauthorized modifications, performing daily and hourly backups, integrating existing POS and inventory management systems and showing analytics dashboard on sales performance. The databases included in the cloud-based sales tracking system are Sales Analytic database,

Inventory and Supply Chain database and POS Transaction database. The Inventory and Supply Chain database consists of tables like Supply_Chain, Stock_Replenishment, Inventory_Transactions and Inventory, whereas the POS Transaction database includes tables like Transaction_items, Cashier_Log, Transactions and Payments. The Sales Analytic database has Customer_Analytics, Sales_Report, Product_performance, Sales_Transaction and Store_Performance tables.

In terms of application architecture, the cloud-based sales tracking system updates the sales transactions and generates sales prediction reports to be stored in the Sales Analytic database. Managers can request insights through the customer relationship management system and the Sales Analytic database provides them real-time reports. The Sales Analytic database updates the synchronization service layer on stock levels, which then optimizes the stock levels of both Inventory and Supply Chain database and POS Transaction database. In terms of infrastructure architecture, the Sales Tracking Server, Report Generation Server, CRM server and the Analytics and Synchronization Server are connected to a switch. A firewall is placed between the enterprise service bus and switch to enhance security and filter traffic.

3.0 - Conclusion

This Buyomart Enterprise Architecture project provides a strategic plan to address the fundamental issues in extending its operations to the south of Malaysia. Under the application of the CSVLOD model, the solutions are constructed to support customer experience, strengthen data-driven decision-making capability, and facilitate workflow operations. Among the core initiatives upgrading the point-of-sale system to simplify the checkout process, enhancing the inventory management system for more efficient stock handling, collaborating with local delivery apps to broaden market reach, and implementing cloud-based sales tracking for real-time and precise analytics.

These changes align with Buyomart's goal of modernising its operations and processes to expand its customer base and stay competitive in the Malaysian retail industry. Alongside offering a clear implementation plan, the comprehensive roadmaps, option evaluations, and solution designs ensure that every phase is practical, scalable, and closely aligned with Buyomart's overall business objectives. Buyomart may achieve smooth operations across all branches and promote long-term growth by implementing these tactics.

In short, this Enterprise Architecture strategy not only addresses the immediate needs of system enhancement and business expansion but also lays a solid foundation for Buyomart's long term digital transformation. It empowers Buyomart to adapt to future retail trends, remain competitive in the market, and deliver consistent as well as high quality service to its customers across all platforms and regions.

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